

# Neighborhood repartition-based oversampling algorithm for multiclass imbalanced data with label noise

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## ABSTRACT

The classification of imbalanced data remains one of the most significant topics in contemporary data analysis. Existing classification algorithms tend to favor majority classes, leading to false predictions and difficulties in addressing class overlap or label noise. These challenges are particularly evident in multiclass settings, where the mutual imbalance relationships among classes become more complex. Despite this, the vast majority of research in this field has concentrated on binary problems, while the more difficult multiclass problems are relatively underexplored. In this paper, we propose a novel data-sampling technique, a *Multiclass Neighborhood Repartition-based Oversampling* (MC-NRO) algorithm. The innovation of this method lies in it considers local data characteristics of each class to constrain the oversampled neighborhood. MC-NRO calculates the mutual potential of different classes to precisely optimize the subregions for generating new instances. By selecting different repartition neighborhoods to meet the needs of specific domain, it can detect outliers and label noise, expand the decision boundary of minority class, and avoid class overlap through data cleaning. The experimental results demonstrate that MC-NRO outperforms other advanced oversampling strategies, ranking first on average across the three evaluation metrics, and exhibits robustness, especially in datasets with high noise levels. More importantly, MC-NRO is highly versatile and can be flexibly applied to various classifiers, and is particularly suitable for processing naturally complex (i.e., not affected by noise) datasets.

## 1. Introduction

Class imbalance problem arises when the number of observations in one class (the majority class) significantly surpasses the number of observations in another class (the minority class). This problem is common in various application scenarios, such as medical diagnosis [1,2], fault detection [3], software defects [4] and financial investment [5]. The crux of solving this problem lies in accurately identifying minority classes. Traditional classification algorithms tend to prefer the majority class, accompanied by a decline in the predictive capabilities for the minority class [6]. However, the imbalance of data itself is not the sole learning difficulty. Imagine that despite a high imbalance ratio (IR), the instances from different classes are well separated. In this case, designing an efficient classifier is not challenging. Importantly, instances from the minority class often come with small clusters of unknown structures, which complicates the learning problem. Small separation, inter-class aggregation, and class overlap distribution greatly increase the learning difficulty of classification

tasks [7]. Therefore, it is necessary to research advanced data preprocessing techniques or innovative classification algorithms to address class imbalance and these associated challenges.

The mentioned challenges in learning from imbalanced data are a focus of in-depth research [8], yet mostly in the context of binary problems [9]. However, the more complex multiclass counterparts may amplify the challenges associated with the imbalanced data [10]. While for binary tasks, the relationships between classes are easy to determine, and in multiclass tasks, the mutual relationships among classes are much more intricate [11]. Popular methods dedicated to binary problems cannot be easily applied to multiclass counterparts, as they fail to model the relationships among classes and the inherited learning difficulties, such as multiple classes overlapping or borderline instances appearing in more than two decision regions. As each class is added, the complexity of class overlap and label noise exponentially increases the difficulty of classification tasks. Although many suggestions decompose the multiclass problem into binary subproblems, this

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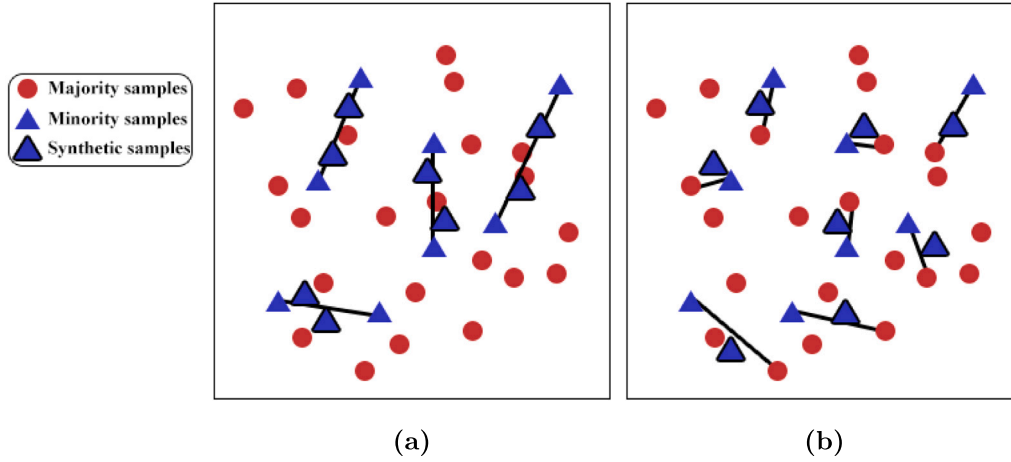


Fig. 1. Visualization of issues in noisy sample generation. (a) Synthetic samples around isolated minorities. (b) Synthetic samples between different classes.

is an oversimplification that loses a valuable perspective on inter class data characteristics [12,13].

Oversampling methods can effectively address binary imbalance problems and are particularly attractive to the multiclass domains, as they work in a preprocessing manner and do not depend on a specific classifier [12]. However, most of oversampling methods are plagued by noisy sample generation, where some noisy minority samples may overlap with majority samples due to the improper generated features [14]. Several issues in existing model designs that could lead to noisy sample generation. Firstly, many oversampling methods generate new instances between a single minority class and its neighbors, which ignores information from other classes and often results in imperfect placement of artificial instances. For instance, as depicted in Fig. 1(a), some isolated minority samples are distanced from their minority neighbors, leading to the generated sample falling into the majority area, thereby increasing class overlap and additional noise. Secondly, some oversampling studies generate samples between the minorities and their majority neighbors to expand the minority regions. In such cases, as shown in Fig. 1(b), synthetic samples fill the empty space between classes, resulting in ambiguous class boundaries. These data-level difficulties become more prominent and critical in multiclass settings [15]. How to deeply explore the distribution differences in data overlapping areas and how to achieve sample expansion without introducing noise, all demand the development of more guided solutions.

To alleviate the above issues, this paper proposes a novel oversampling algorithm named *Multiclass Neighborhood Repartition-based Oversampling* (MC-NRO). It is developed to address imbalanced data with embedded difficulties, that is, atypical data distribution including small separation and class overlap, in the multiclass setting. The advantage of MC-NRO lies in utilizing each instance from every class to estimate class potentials, constraining oversampled areas and guiding the generation of new instances by exploring mutual class potential values. Moreover, it combines the idea of cleaning up minority class neighborhoods with the aim of reducing the impact of class overlap on the performance of classification models. Our methodology deviates from traditional methods based on the use of nearest neighbors to oversample and no longer uses neural networks to generate instances, thus further reducing the drawbacks of existing methods. Thanks to this, we achieve a guided and an intelligent oversampling in multiclass imbalanced situations.

To summarize, the contributions of this study are as follows:

- Proposition of the Multiclass Neighborhood Repartition-based Oversampling algorithm, which guides intelligent oversampling by exploring local data characteristics of each class and is robust to atypical data distributions.
- Application of the radial-based approach to calculate the class potential that repartitions the preset neighborhood after cleaning, which precisely optimizes the placement of new instances and expands the decision boundary of minority classes. MC-NRO has the advantage of partition sampling, and oversampling subregions can be specified to meet specific requirements.
- The computational complexity of MC-NRO is analyzed in detail, and it is shown that it achieves the reliable tradeoff between the preprocessing time and the processing of imbalanced data.

This paper is organized as follows. Section 2 discusses in detail the problem of learning from imbalanced and noisy data and emphasizes the unique characteristics of multiclass problems. Section 3 introduces MC-NRO in detail and extends from the binary classification to the multi-classification, while Section 4 describes the conducted experimental study. The final section concludes this paper and prospects the future development direction in the field of multiclass imbalanced data preprocessing.

## 2. Learning from imbalanced data

In this section, we discuss the above difficulties, starting with the study of binary imbalanced problems, and extending to more challenging multiclass imbalanced problems and label noise. Class imbalance is mainly studied in the context of binary tasks, yet it may also occur in the multiclass setting.

### 2.1. Binary imbalanced problems

The strategies for addressing class imbalance problems can be divided into the following two categories. Data-level solutions: These methods rebalance the training set through oversampling, undersampling, or mixed sampling. After applying such data preprocessing, the rebalanced data can adapt to any standard classifier. The most popular data-level method is Synthetic Minority Oversampling Technique (SMOTE) [16], which is also considered the cornerstone of most oversampling methods. The traditional SMOTE blindly interpolates between existing minority class samples, and its sampling process assumes that the clusters of minority class from a homogeneous area. Unfortunately, when there is class overlap, noise instances, or atypical data distribution, SMOTE may alter the class distribution or interfere with decision boundaries. We will discuss the limitations of neighborhood-based methods in Section 3.1. Numerous modifications to SMOTE have been proposed in the literature, such as the most familiar Borderline-SMOTE [17], which selects minority samples close to the border to generate new instances. Safe-level SMOTE [18] aims to reduce the risk of synthesizing samples in areas of majority class. PF-SMOTE [19]

attempts to fill areas dominated by the minority class while avoiding interpolating noisy samples. Clustering algorithms are combined with SMOTE to reduce noise. Potential anchoring (PA) [20] adopts K-means algorithm to cluster the entire sample space and generate synthetic instances based on the density measurement of each cluster. Position-aware safe boundary interpolation oversampling (PABIO) [21] employs the CFSFDP algorithm to cluster minorities and generate samples around trapped minorities. Koziarski et al. proposed two novel algorithms specifically for class overlap and noise issues: Combined cleaning and resampling (CCR) [9] and Radial-based oversampling (RBO) [22]. CCR combines cleaning the decision border around minority objects with guided synthetic oversampling. RBO finds regions where the synthetic observation of the minority class could be generated on the basis of imbalanced distribution estimation with radial basis functions.

The second category of methods for handling class imbalance problems is algorithm-level solutions. These methods modify or innovate the model based on the failure mechanism of the selected classifier training in the case of imbalanced data, effectively making it robust to imbalanced class distributions [23]. Popular algorithm-level solutions include splitting criteria in decision trees, kernel functions, or modifying the underlying loss function to make them cost-sensitive. Whereas data-level methods work in a preprocessing manner, algorithm-level methods require the selection of a specific classifier and are limited to the selected model. Nevertheless, in many cases, they have been reported to outperform many data-level methods [12].

Recent studies point out that the IR is not the sole source of learning difficulties in imbalanced data. The numerous difficult factors embedded in the nature of data pose additional challenges in the learning process. Borderline and class overlap limit the capacity to achieve a good trade-off between the performance of minority and majority classes. Noisy samples may introduce incorrect information, leading to performance degradation and increase overfitting risk, thereby reducing the model's generalization capabilities.

## 2.2. Multiclass imbalanced problems

Multiclass imbalanced problems pose greater challenges than their binary counterparts due to the more complex relationships among classes [11]. The division of majority and minority classes is no longer so distinct. One of the earlier proposals for multiclass imbalanced taxonomy used the concepts of "multiminority" (a single majority class and multiple minority classes) and "multimajority" (a single minority class and multiple majority classes) [11]. However, this definition is very restrictive, as in reality, the complexity of relationships among classes cannot be clearly defined like this. A single class may be a majority to some, a minority to others, or roughly balanced with the rest. Such situation can cause significant learning difficulties. More importantly, the difficulties in the nature of imbalanced data are more obvious and prevalent in multiclass settings. Multiple classes may cause simultaneous overlap, and the presence of noise and outliers, making such complex regions particularly challenging for any classifier. Furthermore, any class processing must be analyzed for its effect on other classes. All of the above allow us to conclude that it is crucial to design new and specialized methods to address multiclass imbalanced datasets embedded in various learning difficulties.

Taxonomy for addressing multiclass imbalance problems can also be divided into two categories, but it differs from the binary one. Although there remains a division between data-level and algorithm-level solutions, the algorithms differ in the way they handle the multiclass nature of the problem:

The first is binarization solutions. These methods decompose a multiclass problem into  $M(M-1)/2$  (one-versus-one, OVO), or  $M$  (one-versus-all, OVA) binary subproblems [24]. Each subproblem can directly apply existing algorithms [25]. However, binarization solutions

also come at a huge cost, as it loses valuable perspectives on information among the remaining classes. In terms of computational complexity, when the number of classes increases, the number of OVO classifiers grows quadratically, making it impractical for large scale data processing. In addition, for OVA, merging other classes outside of the targeted ones further strengthens class imbalance, which makes it unwise for classes with a high IR. From the aspect of multiclass decomposition strategy in the sampling process, both Multiclass radial-based oversampling (MC-RBO) [26] and Multiclass combined cleaning and resampling (MC-CCR) [8] introduce one-versus-sampling (OVS): sampling the classes except the class being oversampled in an even proportion to form a combined majority class. In each binary subproblem, the majority class is formed by sampling observations from larger classes, which can reduce information loss. Moreover, Noise-robust oversampling (NROMM) [27] proposes combining the original and new synthetic samples from classes with a greater number of observations as the majority (one-versus-ensemble, OVE). This concept resembles the OVA decomposition.

The second is ad hoc solutions. These methods start from treating multiclass problems themselves and propose specialized solutions that utilize non pairwise relationships among individual classes. They require extensive modification to existing algorithms, or explore a novel perspective to overcome data-level imbalance. The cost of complex algorithm design and development is often compensated by excellent performance, as ad hoc solutions are closer to the core problem of multiclass imbalance. Most popular data-level approaches include extensions of SMOTE algorithm to multiclass settings [14,28], combination with feature selection [29], and alternative methods for instance generation by using Mahalanobis distance [30]. SMOTE combined with an iterative partition filter (SMOTE-IPF) [14] overcomes problems caused by noise and borderline examples. Mahalanobis distance oversampling (MDO) [30] intelligently generates samples along the probability contours. Synthetic minority oversampling technique based on k-NN (SMOM) [28] assigns a selection weight to each neighbor direction. Algorithm-level solutions include decision tree adaptations, cost-sensitive learning, and ensemble solutions utilizing bagging [31] and boosting [32].

While the above methods alleviate the class imbalance problem in more complicated situations, they still have limitations on addressing the imbalance problem for multiclass and mixed-type data. To further explain oversampling methods, we compare them on the basis of technical mechanisms and application scenarios in Table 1. MC-RBO [26] roots the instance generation process in the vicinity of the current minority but lacks a cleaning strategy, resulting in unclear class boundaries between minority and majority classes. MC-CCR [8] proposes an energy-based approach to model the optimal radius of the sphere for each minority sample while cleaning the minority neighborhood. However, the sphere expands at a constant rate in each direction, completely ignoring the exact positions of the majority neighbors. This can lead to overlap with other classes as majority samples move outside the sphere. NROMM [27] employs the CFSFDP [33] algorithm to identify candidate minority neighbors and clean up inland samples in overlapped regions after oversampling all classes. Yet, this process also removes some minority instances, resulting in the loss of rare minority data characteristics. Therefore, we propose two cleaning strategy to address the above limitations. Before oversampling, we firstly clean the minority neighborhoods by translation, rather than cleaning the minority classes themselves. This can preserve the characteristics of minority classes to the greatest extent and form a secure authorization area. After all classes oversampled, we perform secondary cleaning on overlapping regions caused by the moved majority samples. Furthermore, different from any of the above methods, the proposed method repartitions the cleaned neighborhood, and then new synthetic instances can be restricted to specific subregions. This not only facilitates more precise authorization for minority classes, but also enhances the detection of outliers and label noise, thereby minimizing the impact of negative incorrect labeling. The detailed process is described in Section 3.2.

**Table 1**

The overall comparison of oversampling algorithms. “Neighborhood-based” measures whether the algorithm uses the nearest neighbors when generating samples. “Cleaning” measures whether the algorithm cleans up the samples that hinder classifiers. “Partition” measures whether to select specific regions for generating new minority samples.

| Algorithms    | Technical mechanisms |          |           | Application scenarios |            |
|---------------|----------------------|----------|-----------|-----------------------|------------|
|               | Neighborhood-based   | Cleaning | Partition | Binary                | Multiclass |
| SMOTE [16]    | ✓                    | ×        | ×         | ✓                     | ×          |
| PF-SMOTE [19] | ✓                    | ×        | ×         | ✓                     | ×          |
| PA [20]       | ×                    | ×        | ×         | ✓                     | ×          |
| PABIO [21]    | ×                    | ×        | ×         | ✓                     | ×          |
| MDO [30]      | ×                    | ×        | ×         | ✓                     | ✓          |
| SMOM [28]     | ✓                    | ×        | ×         | ✓                     | ✓          |
| MC-RBO [26]   | ✓                    | ×        | ×         | ✓                     | OVS [26]   |
| MC-CCR [8]    | ×                    | ✓        | ×         | ✓                     | OVS [8]    |
| NROMM [27]    | ✓                    | ✓        | ×         | ✓                     | OVS [27]   |
| MC-NRO        | ✓                    | ✓        | ✓         | ✓                     | OVS        |

### 2.3. Label noise in the imbalanced problems

High quality labeled data is an important factor in building a high quality classification system. The most serious problem in data preprocessing is the introduction of label noise [34,35]. It has a dual nature. On one hand, it may be the noise caused by human error marks or incorrect measurements when acquiring attributes. On the other hand, it may relate to incorrect data labels. In this study, we aim to examine the robustness of the proposed solution to label noise. This type of noise occurs whenever an observation is assigned an incorrect label [34], and it may lead to the formation of contradictory instances whenever duplicate observations are assigned different class labels [35].

While some literatures and surveys have highlighted this issue [35, 36], relatively few papers have focused on the impact of label noise on the classification prediction performance of imbalanced datasets. Let us primarily consider where the labels come from. The most common approach is to obtain labels from human experts. However, human experts are subjective in labeling, so the distribution of incorrect labels is not uniform [37]. Another approach is to obtain labels from non-experts through crowdsourcing, which provides an effective and scalable way to construct label sets. Nevertheless, creating a comprehensive labeling guidelines is challenging, and any incomplete or ambiguity may cause different interpretations of concepts and inconsistent labels, thereby resulting in greater label noise. In addition, label noise may stem from data corruption [38] or data [39]. Natural and malicious label corruption can dramatically degrade the performance of the classifier.

There are many methods to address label noise. One of the most prevalent approaches is data cleaning. An example of this method is the utilization of SMOTE algorithm with cleaning using the Edited Nearest Neighbors (ENN) [40]. However, when the label noise level is high, the ENN approach may remove too many valuable samples, which is common in imbalanced data analysis tasks. Therefore, developing a data cleaning strategy specifically for the inherent characteristics of label noise is extremely important [36]. MC-CCR [8] cleans the minority neighborhoods through translation, completely disregarding the possibility of increasing overlap with other classes near the boundary. NROMM [27] uses the CFSFDP algorithm to clean up inland samples in the overlapped regions, which may remove some minority class instances, resulting in information loss. The proposed method can effectively mitigate these issues. We adopt two data cleaning. Firstly, to clean majority samples in the minority neighborhood before oversampling to form a secure authorization area, which retain the characteristics of minority classes. Secondly, to remove the moved majority samples in the overlapped region to avoid overlapping.

### 3. Neighborhood repartition-based oversampling algorithm

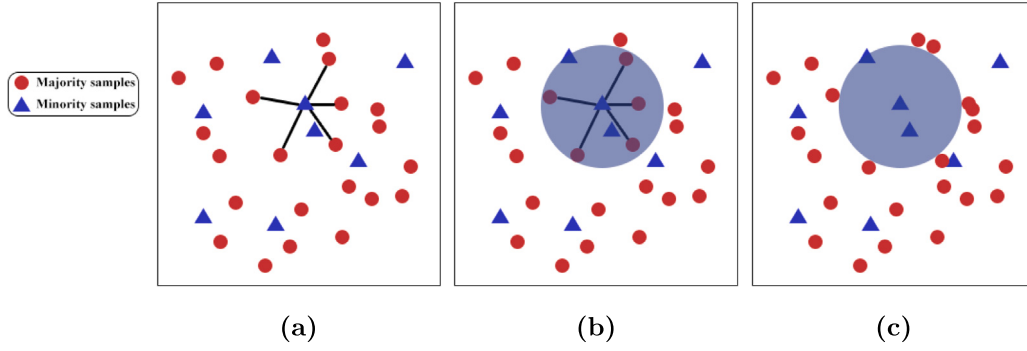
#### 3.1. Limitations and selection of neighborhood-based oversampling strategies

The crux of designing an oversampling algorithm for imbalanced data lies in the selection of regions of interest, that is, the areas for generating new minority samples. Existing neighborhood-based oversampling strategies bring many limitations. For example, Random Oversampling (ROS) duplicates existing minority samples randomly, in this case, the regions of interest are the minority points themselves. This may lead to the model being sensitive to minority samples, thereby increasing overfitting. SMOTE [16] blindly interpolates between the nearest minority neighbors, ignoring the overlap with the majority classes, the regions of interest are located between these minority classes. Although it improves the predictive capabilities of minority classes, centralizing regions of interest in a highly dense majority class observation areas may have a negative impact on identifying majority classes. The main reason causing this behavior is that this series of methods only uses the distribution of minority classes, ignoring the location of majority observations. Finally, in principle, a cleaning procedure, such as SMOTE-ENN [40], could be used to eliminate the overlap, yet in the case of degradation, the newly synthetic samples may still be completely removed in the cleaning step.

Although the severity of neighborhood-based issues can be partially alleviated by reducing the number of nearest neighbors, it cannot eliminate its impact fundamentally. This paper proposes an neighborhood-based selection strategy, that is, only selects the minorities'  $k_1$ -nearest neighbors to form a safe region, contrary to directly generating new samples. Specifically, in the entire sample space, we find the  $k_1$ -majority nearest neighbors for each minority observation, and form a circle using the distance of the  $k_1$ -th point as the radius. This circular neighborhood defines the region of interest for the current minority observation. As a step preceding the oversampling, we suggest cleaning the majority observations located inside the minority neighborhood by moving them outside it. The first process is shown in Fig. 2. If the moved samples form an overlapping set, we will perform a secondary cleaning on the overlapping regions after oversampling, as described in Section 3.2. The aim of such an operation is twofold. Firstly, to bias the classifiers' predictions toward the minority class: by greatly preserving the characteristics of minority classes and forming a secure authorization area for oversampling. Secondly, to reduce the class overlap: by designing the cleaning strategy specifically for the majority observations within the neighborhood.

After the first cleaning stage, a pure neighborhood is designated as the region of interest for oversampling. To further explore these regions, we propose using the size of the calculated neighborhood to weight the selection of minority observations used as the oversampling origin. Analogous to the ADASYN [41], we focus on more difficult





**Fig. 2.** Visualization of the selection of oversampling neighborhood and the first data cleaning with a majority class and a minority class. (a) The original dataset is annotated with 5-majority nearest neighbors of the minority class. (b) Take the distance of the fifth point as the radius to form a neighborhood for the subsequent oversampling. (c) Move the majority observations in the neighborhood outside it.

instances, estimating the difficulty based on the radius of the associated neighborhood. More formally, for a minority point denoted by  $x_i$ , the radius of the associated neighborhood denoted by  $r_i$ , the collection of majority observations denoted by  $\mathcal{X}_{maj}$  and the collection of minority observations denoted by  $\mathcal{X}_{min}$ , we define the number of synthetic samples to be generated nearby  $x_i$  as

$$num_i = \lfloor \frac{r_i^{-1}}{\sum_{k=1}^{|\mathcal{X}_{min}|} r_k^{-1}} \cdot (|\mathcal{X}_{maj}| - |\mathcal{X}_{min}|) \rfloor. \quad (1)$$

This allocation strategy aims to generate more new instances in smaller radius neighborhoods, thereby adaptively shifting the classification decision boundary toward difficult areas. However, different from the ADASYN, the proposed method assigns significant importance to the relative distances of the observations.

### 3.2. Neighborhood repartition and oversampling with radial-basis function

After the above operation, new synthetic minority instances are being sampled within the previously designed cleaning regions. However, due to two considerations: in the first step, it is not clear how many neighbors should be selected, or how this number could be adjusted locally, all of these make the design of neighborhoods more complicated; in addition, similar to the CCR [9], randomly generating new instances may interfere with the majority observations cleaned out by other neighborhoods, and it does not utilize the position information of adjacent minority observations. Therefore, this paper proposes a approach to repartitioning regions of interest within the previous neighborhood, which using the distribution information of mutual classes. The approach is based on the concept of class potential. Intuitively, for any point in space  $x$ , its potential represents the cumulative proximity of that point to its collection of observations  $\mathcal{X}$ . This means if a point is located nearby a large number of observations from the collection, it will be characterized by a higher potential regarding this collection. To model this behavior, we will attach the Gaussian radial basis function (G-RBF) to define the potential function as follows: given a collection of observations  $\mathcal{X}$ , a point in space  $x$ , parameter  $\epsilon$  representing the spread of a single G-RBF, and using the  $L1$  - norm as a distance measure, the potential function is defined as

$$\phi(x, \mathcal{X}, \epsilon) = \sum_{i=1}^{|\mathcal{X}|} e^{-\left(\frac{\|x_i - x\|_1}{\epsilon}\right)^2}. \quad (2)$$

The choice of the parameter  $\epsilon$  affects the range of class potential estimation: a lower  $\epsilon$  will produce potentials concentrated around existing observations, whereas a higher  $\epsilon$  will form smoother potentials. In the case of imbalanced data oversampling, class potential representation carries more class distribution information than the oversampling regions created by neighborhood-based methods, as demonstrated in the RBO [22]. However, unlike RBO, the proposed strategy uses class

potential to repartition previous neighborhoods instead of directly oversampling. This improves the drawbacks of RBO: it roots the instance generation process in the vicinity of the current minority class and cannot optimally expand the decision boundary.

Specifically, candidate samples are randomly generated with a uniform probability in the entire neighborhood, and their potentials are calculated by the G-RBF. The preset neighborhood can be repartitioned by exploring the mutual potentials of different classes. We design a detailed formulation to partition neighborhoods as follows: given a minority point  $x$ , its potential  $\phi^0$ , the collection of potentials of candidates  $U$ , parameter  $n$  representing the number of the neighborhood need to be partitioned, the neighborhood-based repartition function  $R$  is defined as

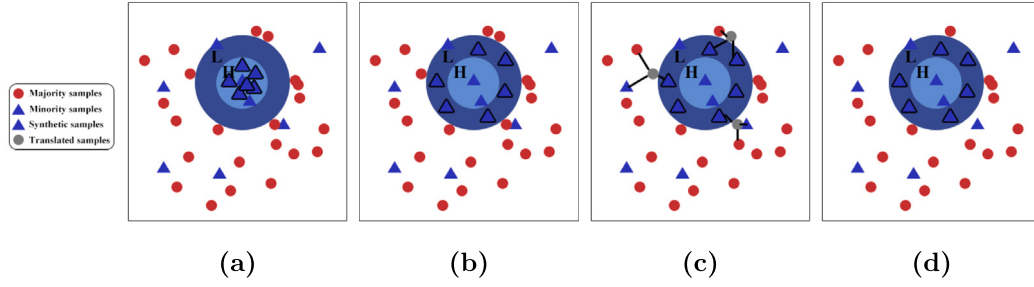
$$R = \phi^0 + \frac{1}{n} (Max(U) - \phi^0). \quad (3)$$

The proposed strategy partitions the designated neighborhood into  $n$  target regions. After partitioning, new synthetic instances can be restricted to a specific subregion. Furthermore, the oversampling region can be partitioned more finely, which provides a flexibility in selecting appropriate sampling areas for complex datasets within a global framework. The preset neighborhood is partitioned into two target regions in Fig. 3. As we can see, sampling in High potential region (Fig. 3(a)), where synthetic samples are closer to the current minority point, closely resembling the characteristic of the original minority observations. This is beneficial to improving the classifier's predictive performance for minority classes while minimizing the impact on correctly identifying majority classes. It can also better detect outliers and label noise, reducing the negative impact of incorrect labeling results. The opposite is sampling in Low potential region (Fig. 3(b)), where synthetic samples are closer to the moved majority observations outside their neighborhoods, effectively expanding the decision boundary of minority classes.

It is worth mentioning that we clean the majority observations located inside the minority neighborhood by moving them outside it, which may overlap with majority samples cleaned by other neighborhoods or cause unclear class boundaries, as described in Section 3.1. Therefore, we recommend cleaning the moved majority observations near the boundary twice after all classes oversampled. Starting with a moved majority sample  $x_{moved} \in X^c$ , if its  $k_2$ -nearest neighbor set  $N(x_{moved})$  includes samples from the other classes  $\bar{X}^c$  (Fig. 3(c)), an overlapping region is formed. Then we remove the current majority sample in the overlapping region to clarify the class boundary (Fig. 3(d)).

### 3.3. Binary neighborhood repartition-based oversampling

In the binary setting, the neighborhood repartition strategy use class potential to guide oversampling in various ways. Partition sampling can precisely optimize specific subregions for generating synthetic



**Fig. 3.** Visualization of repartitioning the previous neighborhood based on the class potential and oversampling in the specific region. (a) Sampling in High (H) region. (b) Sampling in Low (L) region. (c) Judging whether moved majority observations cause overlapping after all classes oversampled. (d) Removing the moved majority observations in the overlapping region; Ready for classification.

instances. These subregions can be specified as an input parameter to meet the requirements of specific domain or automatically selected through cross-validation.

In this paper, we use an improvement of a simple hill climbing algorithm. Starting with a randomly chosen minority point, we find its  $k$ -majority nearest neighbors to form a neighborhood. Firstly, we clean the minority neighborhood by moving the majority samples outside of it, and use the size of the calculated neighborhood in the process of assigning a sampling ratio to the point. Then, candidate samples are randomly generated in the neighborhood, and their potentials are estimated using the G-RBF to form a collection  $U$ . The designed neighborhood can be repartitioned using the function  $R$ , retaining new synthetic instances within the specific region at its ratio. After oversampled, the second cleaning identifies and removes the moved majority samples in the overlapping region. To reduce the calculational cost of the algorithm, we consider the calculational complexity of class potential estimation: instead of selecting all of the minority observations for calculation, the algorithm only uses  $k$ -nearest neighbors of the minority observation and minimizes the sampling ratios as much as possible. The pseudocode of the *Binary Neighborhood Repartition-based Oversampling* (B-NRO) algorithm is presented in Algorithm 1.

#### Algorithm 1 B-NRO

**Input:** collection of majority observations  $\mathcal{X}_{maj}$  and minority observations  $\mathcal{X}_{min}$ ,  $k_1$ -nearest neighbors used for forming neighborhoods, spread of radial basis function  $\epsilon$ , number of *iterations* used for class potential estimation,  $n$  used for repartitioning the neighborhood,  $k_2$ -nearest neighbors used for cleaning the overlapping region

**Output:** collection of moved majority observations  $\mathcal{X}'_{maj}$  and collection of synthetic minority observations  $Syns$

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1: Function: B-NRO ( $\mathcal{X}_{maj}, \mathcal{X}_{min}, k_1, \epsilon, iterations, n, k_2$ )
2:  $Syns \leftarrow \emptyset$  # synthetic minority observations  $Syns$ 
Step 1: Selecting the oversampling area based on nearest neighbors and cleaning the minority neighborhood
3:  $r \leftarrow$  zero vector of size  $|\mathcal{X}_{min}|$  # radii of neighborhoods associated with the minority observations
4:  $t \leftarrow$  zero matrix of size  $|\mathcal{X}_{maj}| \times m$  # with  $m$  denoting the number of features # move the majority observations outside the neighborhood
5: for each minority observation  $x_i$  in  $\mathcal{X}_{min}$  do
6:   for each majority observation  $x_j$  in  $\mathcal{X}_{maj}$  do
7:      $d_j \leftarrow \|x_i - x_j\|_1$ 
8:   end for
9:   sort  $\mathcal{X}_{maj}$  by  $d$ 
10:  find  $k_1$ -nearest neighbors from  $\mathcal{X}_{maj}$  for the current minority observation
11:   $r_i \leftarrow d_{k_1}$  # take the distance of the  $k_1$ -th point as the radius
12:  for each majority observation  $x_j$  in  $\mathcal{X}_{maj}$  do
13:    if  $d_j < r_i$  then
14:       $t_j \leftarrow t_j + \frac{r_i - d_j}{d_i} \cdot (x_j - x_i)$ 

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15:   end if
16: end for
17: end for
18:  $\mathcal{X}'_{maj} \leftarrow \mathcal{X}_{maj} + t$  # collection of moved majority observations  $\mathcal{X}'_{maj}$ 
Step 2: Repartitioning the previous neighborhood with Radial-basis function
19: for each minority observation  $x_i$  in  $\mathcal{X}_{min}$  do
20:    $num_i = \lfloor \frac{r_i^{-1}}{\sum_{k=1}^{|\mathcal{X}_{min}|} r_k^{-1}} \cdot (|\mathcal{X}_{maj}| - |\mathcal{X}_{min}|) \rfloor$ 
21:    $\mathcal{X}^i_{min} \leftarrow$  collection of minority observations in the neighborhood
22:    $\phi^0 \leftarrow \phi((x_i, \mathcal{X}^i_{min}, \epsilon))$ 
23:    $X^* \leftarrow \emptyset$  # collection of candidate samples
24:    $U \leftarrow \emptyset$  # collection of candidate potentials
25:   for  $i \leftarrow 1$  to iterations do
26:      $x_i^* \leftarrow$  random samples inside the neighborhood with radius  $r_i$ 
27:      $U_i \leftarrow \phi((x_i^*, \mathcal{X}^i_{min}, \epsilon))$ 
28:   end for
29:    $R = \phi^0 + \frac{1}{n} (Max(U) - \phi^0)$  # estimate potential bounds to repartition the neighborhood
30:   for  $i \leftarrow 1$  to iterations do
31:     if  $U_i \geq R$  then
32:        $temp_i \leftarrow x_i^*$  # retain suitable samples in the region  $R$ 
33:     end if
34:   end for
35:   if  $Len(temp_i) \geq num_i$  then
36:      $Syns_i \leftarrow temp_i(Len(num_i))$  # retain  $num_i$  suitable samples
37:   end if
38:    $Syns_i \leftarrow Syns_i \cup x_i$ 
39: end for
40:  $Syns \leftarrow Syns_i$ 
Step 3: Cleaning the moved majority samples in the overlapping region
41: for each moved majority observation  $x_j$  in  $\mathcal{X}'_{maj}$  do
42:    $N(x_{moved}) \leftarrow$  its  $k_2$ -nearest neighbors in the entire sample space
43:   if  $x_j \notin X^c$  then
44:     remove  $x_j$  in  $\mathcal{X}'_{maj}$  # remove  $x_j$  in the overlapping region
45:   end if
46: end for
47: return  $\mathcal{X}'_{maj}, Syns$ 

```

It is worth discussing that how B-NRO compares to several popular oversampling methods, based on a highly binary imbalanced dataset, the detailed differences are shown in Fig. 4. As can be seen, compared to SMOTE-based methods, B-NRO tends to introduce less class overlap. SMOTE [16] is greatly affected by disjoint data distribution and label noise, while Borderline-SMOTE [17] is only more sensitive to boundary samples. RBO [22] generates very conservative and densely clustered samples, which may increase overfitting in a manner similar to random oversampling. CCR [9] adopts a cleaning strategy by moving majority observations outside the sphere, yet random synthetic samples

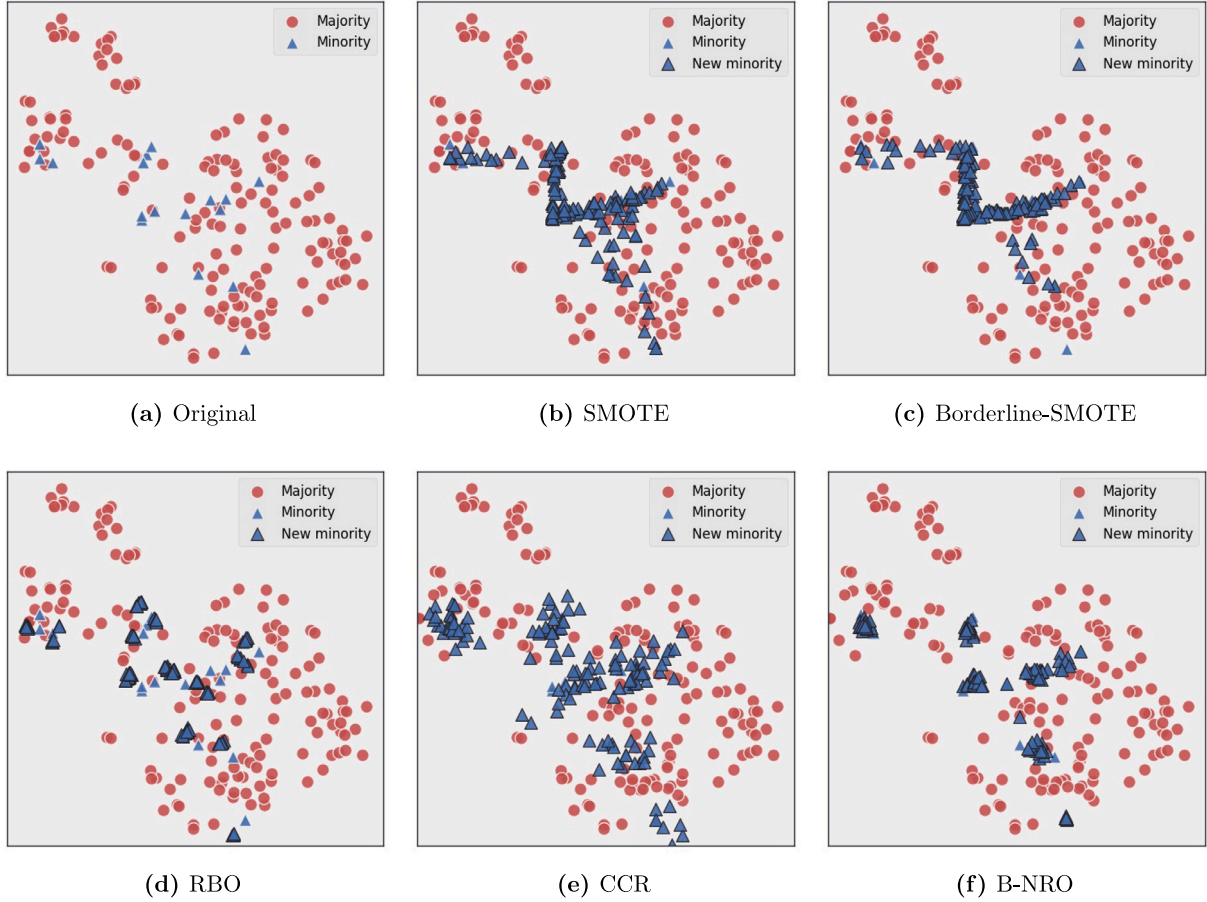


Fig. 4. Illustration of differences between B-NRO and different oversampling algorithms on binary imbalanced data with a majority class (red circle) and a minority class (blue triangle).

within the sphere may interfere with the decision boundary. B-NRO repartitions the neighborhood to retain synthetic instances within the specific area while cleaning majority samples near boundaries. This can enhance bias towards the minority class distribution while minimizing class overlap. Furthermore, B-NRO retains more constrained samples based on the underlying potential, which can avoid overfitting risks with larger regions of interest.

### 3.4. Multiclass neighborhood repartition-based oversampling

To extend the B-NRO algorithm to the multiclass setting, we introduce a decomposition iteration strategy that oversamples only one class at a time. First of all, sort the classes by the number of associated observations in descending order. For each class except the unique majority class, we combine a collection of majority classes by aggregating a fraction of the sample from each of the previously considered classes. Then, we apply the B-NRO algorithm, treating the current class as the minority and the combined majority observations as the majority. New synthetic minority instances and the moved majority observations are integrated into the original dataset for subsequent combinations. Finally, different from B-NRO, after oversampling all classes, we clean the moved majority samples near boundaries to reduce overlap between classes. The pseudocode of the *Multiclass Neighborhood Repartition-based Oversampling* (MC-NRO) algorithm is presented in Algorithm 2. Compared to an alternative OVA decomposition strategy, this sampling strategy exhibits two advantages. First, it typically reduces the calculational cost, as the number of combined majority observations is often smaller than the entire sample space. Second, it assigns equal weight to each class in the process of combining the majority, while in OVA, classes with more observations could dominate the remaining classes.

Our experiments indicate that using the OVS decomposition strategy generally achieves better classification performance.

#### Algorithm 2 MC-NRO

**Input:** collections of all observations  $\mathcal{X}$ , with  $\mathcal{X}^c$  denoting a subcollection of observations belonging to class  $c$ ,  $k_1$ -nearest neighbors used for forming neighborhoods, spread of radial basis function  $\epsilon$ , number of iterations used for class potential estimation,  $n$  used for repartitioning the neighborhood,  $k_2$ -nearest neighbors used for cleaning the overlapping region

**Output:** collections  $Syns$ , with  $Syns_i$  denoting a collection of synthetic minority observations generated for  $i$ th class

- 1: **Function:** MC-NRO ( $\mathcal{X}, k_1, \epsilon, iterations, n, k_2$ )
- 2:  $Syns \leftarrow \emptyset$  # synthetic minority observations  $Syns$
- 3:  $C \leftarrow$  collections of all classes, sorted by the number of all classes in descending order
- 4: **for** each class  $i$  in  $|C|$  **do**
- 5:    $n_c \leftarrow$  number of classes with higher number of observations than  $C_i$
- 6:   **if**  $n_c = 0$  **then**
- 7:      $Syns_i \leftarrow \emptyset$
- 8:   **else**
- 9:      $\mathcal{X}_{min} \leftarrow \mathcal{X}^{C_i}$
- 10:     $\mathcal{X}_{maj} \leftarrow \emptyset$
- 11:    **for**  $j \leftarrow 1$  to  $n_c$  **do**
- 12:     add  $\lfloor \frac{|\mathcal{X}^{C_1}|}{n_c} \rfloor$  randomly choose observations from  $\mathcal{X}^j \cup Syns_j$  to  $\mathcal{X}_{maj}$
- 13:    **end for**

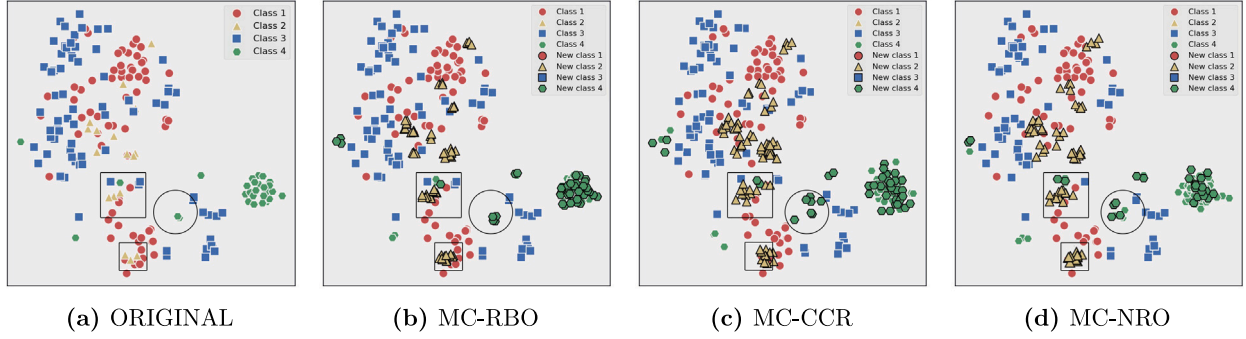


Fig. 5. Illustration of different oversampling algorithms on a multiclass dataset with four classes.

```

14:  $\mathcal{X}'_{maj(i)}, \text{Syn}_{s_i} \leftarrow \text{B-NRO}(\mathcal{X}_{maj}, \mathcal{X}_{min}, k_1, \epsilon, \text{iterations}, n, k_2)$  with-
out Step 3
15: end if
16: end for
17: for each moved majority observation  $x_j$  in  $\mathcal{X}'_{maj}$  do
18:    $N(x_{moved}) \leftarrow$  its  $k_2$ -nearest neighbors in the entire sample space
19:   if  $x_j \notin X^c$  then
20:     remove  $x_j$  in  $\mathcal{X}'_{maj}$  # remove  $x_j$  in the overlapping region
21:   end if
22: end for
23: return  $\text{Syn}_{s_i}$ 

```

An illustration comparing the behavior of the MC-NRO algorithm with other latest multiclass algorithms is presented in Fig. 5. Unlike MC-RBO [26], MC-NRO is more resilient to outliers and label noise. MC-RBO roots the oversampling process on the minority class itself. In any case, synthetic samples are located in a close proximity to the original observations. However, due to the lack of cleaning strategies, it is still unclear whether it will overlap with clusters from other classes, as shown in the square part of Fig. 5. While MC-CCR [8] reduces the negative impact of class overlap by cleaning the oversampling regions, synthetic instances are placed freely and randomly. In addition, the cleaned majority class may confuse decision boundaries, as shown in the circular part of Fig. 5. MC-NRO alleviates these two shortcomings: firstly, partition sampling meets specific needs. We use the neighborhood repartition function to extend oversampling regions and precisely position the subregions of synthetic samples. This approach can reconstruct the feature space of minority classes and provide more targeted authorization for minority classes. Secondly, data cleaning forms a secure authorization area. We identify the moved majority samples near the boundary and removes them in the overlap region. This can ignore outliers surrounded by majority observations and reduce overlap between classes. All this extension can be adaptable to the different characteristics of different datasets.

### 3.5. Complexity analysis

In the experiment, we define  $n$  as the total number of observations,  $n_{maj}$  and  $n_{min}$  as the number of observations of the majority and minority, respectively,  $m$  as the number of features,  $c$  the number of classes in the multiclass,  $k$  the number of selected  $k$ -nearest neighbors, and  $i$  the number of iterations to generate new samples for each minority class. Consider the worst-case complexity of the B-NRO algorithm. The algorithm can be divided into four steps: determining the neighborhood radius, cleaning the majority observations in the neighborhood, synthesizing new samples and redividing the neighborhood. Each step above is applied iteratively to every minority observation. After all classes are oversampled, consider the worst-case complexity of the MC-NRO algorithm in cleaning the moved majority observations in the overlapping region.

The first step is to determine the neighborhood radius. (a) Calculate the distance, which represents the distance between each minority class and all the majority classes. It requires  $m$  and  $n_{maj}$ , and the complexity of each calculation is  $o(m)$ . The computational complexity of the combination is  $o(mn_{maj})$ . (b) Sort the above distances in ascending order. The computational complexity of each is  $O(n_{maj} \log n_{maj})$ . Select the distance of the  $k$ th point as the radius. The combined complexity of the above operations is equal to  $o(mn_{maj} + n_{maj} \log n_{maj})$ , which is the complexity of determining the neighborhood radius for one minority class. The computational complexity for all minority classes is equal to  $o((mn_{maj} + n_{maj} \log n_{maj}) n_{min})$ , which is simplified as  $o((m + \log n)n^2)$ .

The second step is to clean up majority observations in the neighborhood. It is necessary to calculate the translation distance of each majority sample in the neighborhood. Suppose that in the worst case, it is necessary to clean up  $n_{maj}$  with  $m$  feature, with a complexity of  $o(mn_{maj})$ , and simplified as  $o(mn)$ .

The third step is to synthesize new samples. (a) Calculate the number of new samples needed to generate for each minority observation. The complexity is  $o(n_{min})$ . With respect to the function  $\phi$  of the minority in the neighborhood, the calculation complexity is  $mn_{min}$ . (b) Randomly oversample in the neighborhood through iteration. The complexity of each new sample is  $o(m)$ . The combined calculation complexity is  $o(im)$ . With respect to the function  $\phi$  of the minority in the neighborhood, the calculation complexity is  $o(imn_{min})$ .

The fourth step is to redivide the neighborhood, (a) judge whether the randomly oversampled samples are in the selected area, and the computational complexity is  $o(i)$ , (b) judge whether the number of samples in the selected area is larger than the specified number of generated samples, if the new samples are randomly selected in the specified number, the computational complexity is  $o(i)$ , and the combined computational complexity of all minority observations is  $o((mn_{min} + imn_{min} + i + i)n_{min})$ , which is simplified as  $o(imn^2)$ .

As can be seen, it is assumed that in the worst case, the computational complexity of the B-NRO algorithm is equal to  $o((im + \log n)n^2)$ . Since the calculation cost of the multiple classification algorithm is related to the number of incoming binary classification classes, and does not depend on the number of subsequent oversampling observations, before all classes are oversampled, the worst-case calculation complexity of the MC-NRO is equal to  $o(c(im + \log n)n^2)$ .

Finally, clean the moved majority observations in the overlapping region in MC-NRO. Suppose that in the worst case, it is necessary to clean up  $n_{maj}$  with  $m$  feature, with a complexity of  $o(mn_{maj})$ , and simplified as  $o(mn)$ . Therefore, the worst-case calculation complexity of the MC-NRO is equal to  $o(c(im + \log n)n^2)$ .

## 4. Experimental study

In this section, we will describe the details of experimental studies to evaluate the practicality of the proposed MC-NRO. The research questions raised in this paper are as follows:



**Table 2**  
Details of the multiclass imbalanced datasets used in the experiments.

| Datasets        | #N   | #C | #F <sub>N</sub> | #F <sub>C</sub> | IR    | Class distribution               |
|-----------------|------|----|-----------------|-----------------|-------|----------------------------------|
| Automobile      | 159  | 6  | 15              | 10              | 16    | 3/20/48/46/29/13                 |
| Balance         | 625  | 3  | 4               | 0               | 5.88  | 49/288/288                       |
| Car             | 1728 | 4  | 0               | 6               | 18.62 | 384/69/1210/65                   |
| Cleveland       | 297  | 5  | 13              | 0               | 12.31 | 160/54/35/35/13                  |
| Contraceptive   | 1473 | 3  | 6               | 3               | 1.89  | 629/333/511                      |
| Dermatology     | 358  | 6  | 34              | 0               | 5.55  | 111/60/71/48/48/20               |
| Ecoli           | 336  | 8  | 7               | 0               | 71.5  | 143/77/2/2/35/20/5/52            |
| Flare           | 1066 | 6  | 0               | 11              | 7.7   | 147/211/239/95/43/331            |
| Glass           | 214  | 6  | 9               | 0               | 8.44  | 70/76/17/13/9/29                 |
| Hayes-roth      | 160  | 3  | 4               | 0               | 2.1   | 65/64/31                         |
| Led7digit       | 500  | 10 | 0               | 7               | 1.54  | 45/37/51/57/52/52/47/57/53/49    |
| Lymphography    | 148  | 4  | 3               | 15              | 40.5  | 4/61/81/2                        |
| New-thyroid     | 215  | 3  | 5               | 0               | 5     | 150/35/30                        |
| Page-blocks     | 548  | 5  | 10              | 0               | 164   | 492/33/3/8/12                    |
| Thyroid         | 720  | 3  | 6               | 15              | 39.18 | 17/37/666                        |
| Vehicle         | 846  | 4  | 18              | 0               | 1.1   | 218/212/217/199                  |
| Wine            | 178  | 3  | 13              | 0               | 1.48  | 59/71/48                         |
| Winequality-red | 1599 | 6  | 11              | 0               | 68.1  | 10/53/681/638/199/18             |
| Yeast           | 1484 | 10 | 8               | 0               | 92.6  | 463/5/35/44/51/163/244/429/20/30 |
| Zoo             | 101  | 7  | 0               | 16              | 10.25 | 41/20/5/13/4/8/10                |

Notes: #N denotes the number of instances; #C denotes the number of classes; #F<sub>N</sub> denotes the number of numerical variables; #F<sub>C</sub> denotes the number of categorical variables.

**Q1:** What are the optimal parameters setting for MC-NRO, and how they affect the behavior of the algorithm?

**Q2:** What is the prediction performance of MC-NRO in comparison to the state-of-the-art oversampling methods?

**Q3:** How flexible is MC-NRO be effectively used for various different classifiers?

**Q4:** How robust is MC-NRO in dealing with label noise?

#### 4.1. Preparation

**Multiclass imbalanced datasets.** In this paper, we choose 20 multiclass imbalanced datasets from KEEL repository as the benchmark [42]. Their details are presented in Table 2. The selection of datasets is based on our previous research on instance level difficulties in multiclass imbalanced datasets, which embed many challenging characteristics, such as intra-class separation, class overlap, frequent borderline, and noise instances [15]. The experiment specifically uses inverse frequency weighted One-hot encoding [43] for categorical variables.

**Reference oversampling methods.** The proposed MC-NRO is compared with the state-of-the-art oversampling algorithms for multiclass imbalanced data: SMOTE using the round-robin decomposition strategy (SMOTE-all), SMOTE combined with an iterative partition filter (SMOTE-IPF) [14], Mahalanobis distance oversampling (MDO) [30], Multiclass synthetic minority oversampling (SMOM) [28], Multiclass radial-based oversampling (MC-RBO) [26], Multiclass combined cleaning and resampling (MC-CCR) [8], and Noise-robust oversampling (NROMM) [27]. Parameters of the reference methods used in the entire experiments are presented in Table 3.

**Base classifiers.** To verify the validity and flexibility of MC-NRO, we evaluate its effects with four different base classifiers: decision tree (C5.0), neural network (multi-layer perceptron with backpropagation, MLP), lazy learner (k-nearest neighbors, k-NN), and probabilistic classifier (Naive Bayes, NB). Detailed parameter ranges for used classifier algorithms are presented in Table 3.

**Evaluation procedure.** For training and testing the classifiers, we apply a stratified 2-fold cross-validation, repeated 5 times with the final performance averaged over 10 results. Parameters are selected independently for each data partition using an internal 3-fold cross-validation on the training data.

**Evaluation metrics.** We assess the performance of classifiers on multiclass imbalanced data using three dedicated metrics [44]: Average Accuracy (AvAcc), Class Balance Accuracy (CBA), and multiclass

**Table 3**  
Parameters of the classifiers and the sampling algorithms used in the experiments.

| Algorithm      | Parameters                                                                                                                                                                                                                                                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KNN [26]       | k-nearest neighbors $\in [1, 3, \dots, 11]$                                                                                                                                                                                                                                 |
| MLP [26]       | training: rprop;<br>iterations $\in [100, 200, \dots, 1000]$ ;<br>#hidden neurons = $\frac{\#input + \#output}{2}$                                                                                                                                                          |
| SMOTE-all [16] | k-nearest neighbors = 5;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$                                                                                                                                                                                           |
| SMOTE-IPF [14] | $n = 9$ ;<br>k-nearest neighbors = 5;<br>$n$ partitions = 9;<br>$k$ iterations = 3;<br>$p = 0.01$ ;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$                                                                                                                |
| MDO [30]       | $K1 \in [1, 2, \dots, 10]$ ; $K2 \in [2, 4, \dots, 20]$ ;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$                                                                                                                                                          |
| SMOM [28]      | $K1 \in [2, 4, \dots, 20]$ ; $K2 \in [1, 2, \dots, 10]$ ;<br>$nTh \in [1, 2, \dots, 10]$ ;<br>$rTh, w_1, w_2, r_1, r_2 \in [0.1, 0.2, \dots, 1]$ ;<br>$k_{max} = \max(k_1, k_2)$ ;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$                                 |
| MC-RBO [26]    | $\gamma \in [0.001, 0.01, \dots, 10]$ ;<br>step $\in [0.001, 0.005, \dots, 0.1]$ ;<br>iterations $\in [1000, 2000, \dots, 8000]$ ;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$ ;<br>neighbors for potential approximations $\in [10, 25, 50, 100]$             |
| MC-CCR [8]     | energy $\in [0.001, 0.0025, 0.005, 0.01, \dots, 100]$ ;<br>cleaning strategy: translation;<br>selection strategy: proportional;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$                                                                                    |
| NROMM [27]     | $d_c \in [2\%, 3\%, \dots, 10\%]$ ;<br>$k_1, k_2, k_3, k_4, k_5, k_6$ -nearest neighbors $\in [1, 2, \dots, 10]$ ;<br>$\alpha, \beta \in [1, 1.5, 2, 2.5, \dots, 10]$ ;<br>$\eta \in [0.001, 0.005, \dots, 0.5]$ ;<br>oversampling ratio $\theta \in [50, 100, \dots, 500]$ |
| MC-NRO         | $k_1, k_2$ -nearest neighbors $\in [1, 2, \dots, 10]$ ;<br>$\epsilon \in [0.001, 0.01, \dots, 10]$ ;<br>iterations $\in [500, 1000, \dots, 8000]$ ;<br>oversampling ratio = $[100, 200, \dots, 1000]$ ;<br>$n \in [1, 2, \dots, 10]$                                        |

G-measure ( $mGM$ ). They are expressed as follows:

$$AvAcc = \frac{\sum_{i=1}^M TPR_i}{M} \quad (4)$$

$$CBA = \frac{\sum_{i=1}^M \frac{mat_{i,i}}{\max(\sum_{j=1}^M mat_{i,j}, \sum_{j=1}^M mat_{j,i})}}{M} \quad (5)$$

**Table 4**  
Impact of MC-NRO parameters on three evaluation metrics and four base classifiers.

| Parameters |           |               | C5.0         |              |              | MLP          |              |              | KNN          |              |              | NB           |              |              |
|------------|-----------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Cleaning   | Selection | Decomposition | AvAcc        | CBA          | mGM          | AvAcc        | CBA          | mGM          | AvAcc        | CBA          | mGM          | AvAcc        | CBA          | mGM          |
| T+R        | P         | S             | <b>80.23</b> | <b>77.73</b> | <b>77.56</b> | <b>79.48</b> | <b>78.33</b> | <b>77.55</b> | <b>77.81</b> | <b>75.91</b> | <b>74.51</b> | <b>78.90</b> | <b>76.43</b> | <b>76.36</b> |
| T+R        | P         | A             | 78.29        | 77.26        | 76.94        | 78.55        | 77.48        | 77.21        | 76.44        | 75.23        | 72.63        | 77.09        | 75.52        | 75.04        |
| T+R        | R         | S             | 73.55        | 72.62        | 70.35        | 73.08        | 72.92        | 71.26        | 70.76        | 69.08        | 67.85        | 73.82        | 70.00        | 68.79        |
| T+R        | R         | A             | 72.49        | 70.85        | 69.52        | 72.18        | 72.22        | 69.05        | 70.60        | 68.2         | 65.62        | 71.94        | 68.88        | 67.29        |
| T          | P         | S             | 79.54        | 76.06        | 75.72        | 78.85        | 78.39        | 75.22        | 76.75        | 75.86        | 73.87        | 76.42        | 75.26        | 74.43        |
| T          | P         | A             | 77.89        | 75.58        | 75.88        | 77.78        | 77.09        | 74.82        | 75.35        | 74.98        | 72.80        | 74.71        | 73.88        | 73.00        |
| T          | R         | S             | 73.13        | 71.11        | 68.25        | 71.77        | 70.56        | 68.63        | 70.14        | 68.85        | 67.90        | 71.52        | 68.58        | 66.73        |
| T          | R         | A             | 71.95        | 69.35        | 67.42        | 71.34        | 70.27        | 66.42        | 69.97        | 68.41        | 65.20        | 69.42        | 67.14        | 65.54        |
| R          | P         | S             | 74.72        | 71.33        | 71.76        | 73.51        | 72.27        | 69.53        | 71.13        | 71.53        | 70.00        | 72.60        | 71.93        | 70.39        |
| R          | P         | A             | 73.38        | 70.58        | 71.38        | 71.56        | 69.40        | 67.78        | 68.93        | 67.90        | 66.35        | 71.97        | 69.69        | 68.82        |
| R          | R         | S             | 70.85        | 67.28        | 68.59        | 69.81        | 68.57        | 66.72        | 67.45        | 66.39        | 65.27        | 70.09        | 68.59        | 67.73        |
| R          | R         | A             | 68.44        | 65.15        | 67.25        | 68.02        | 66.42        | 64.75        | 65.41        | 64.78        | 63.22        | 65.54        | 64.78        | 64.59        |

$$mGM = \sqrt[M]{\prod_{i=1}^M recall_i} \quad (6)$$

Where,  $M$  denotes the number of classes,  $TPR$  denotes the proportion of all predicted positive classes to all true positive classes, and  $mat_{i,j}$  denotes the number of instances of the true class  $i$ , which were predicted as class  $j$ . It is also worth mentioning that the performance evaluation metrics for imbalanced data classification is still an open problem.

**Statistical analysis.** To evaluate the statistical significance of the experimental results, we employ a combined 10-fold cross-validation F-test [45] for pairwise comparison on a single dataset, a Friedman ranking test with Shaffer post-hoc analysis [46] for comparison of methods over multiple datasets. The results of all employed tests are reported at significance level  $\alpha = 0.05$ .

**Implementation and reproducibility.** The proposed MC-NRO algorithm was implemented in Python programming language and published as an open-source code at.<sup>1</sup>

#### 4.2. Examination of a validity of the extension designs

First, we examine the validity of the design choices in our MC-NRO algorithm within the multiclass setting. Specifically, three variable parts of MC-NRO that can affect its performance can be distinguished. The first variable is the cleaning strategy, which refers to the way MC-NRO handles majority instances inside the designated neighborhood. The proposed approach to handling these instances firstly moves them outside the neighborhood (translation, T), and then removes the moved majority observations in the overlapped region (removal, R), which is compared with simple translation or complete removal. The second variable is the selection strategy, the way of assigning greater sampling ratio for every minority observation with small associated neighborhood radius. The sampling ratio is inversely proportional to the radius of the neighborhood (proportional, P), which is compared with randomly generated samples, with no associated weight (random, R). The last variable is the decomposition strategy, comparing the OVS decomposition (sampling, S) with OVA decomposition (all, A).

To validate the design choices experimentally, we compare all possible combinations of the outlined parameters on four classifiers. The average results, based on all considered datasets, are shown in Table 4. MC-NRO selects a combination of parameters, that is cleaning by translation and removal, proportional seed instances selection and the OVS decomposition strategy, which displays the best average performance of all baseline classifiers and evaluation metrics. In particular,

**Table 5**

Shaffer's tests for comparison between  $MC - NRO_{OVS}$  and reference  $MC - NRO_{OVO}$  and  $MC - NRO_{OVA}$  on used base classifiers. We report obtained  $p$ -values. Symbol '>' stands for situation in which MC-NRO is statistically superior and '=' for when there are no significant differences.

| Classifier | Hypothesis           | AcAcc     | CBA       | mGM       |
|------------|----------------------|-----------|-----------|-----------|
| C5.0       | vs. $MC - NRO_{OVO}$ | >(0.0195) | >(0.0349) | >(0.0352) |
|            | vs. $MC - NRO_{OVA}$ | >(0.0287) | >(0.0391) | >(0.0382) |
| MLP        | vs. $MC - NRO_{OVO}$ | =(0.1293) | >(0.0326) | =(0.1416) |
|            | vs. $MC - NRO_{OVA}$ | =(0.1167) | >(0.0256) | =(0.1305) |
| KNN        | vs. $MC - NRO_{OVO}$ | >(0.0265) | >(0.0211) | =(0.0678) |
|            | vs. $MC - NRO_{OVA}$ | >(0.0323) | >(0.0271) | =(0.0919) |
| NB         | vs. $MC - NRO_{OVO}$ | >(0.0328) | >(0.0146) | >(0.0346) |
|            | vs. $MC - NRO_{OVA}$ | >(0.0336) | >(0.0281) | >(0.0462) |

the cleaning strategy plays a significant role in achieving satisfactory results.

For the integrality of the experiment, we also apply the OVO decomposition: considering all the classes in a pairwise manner. Due to space limitations, only the outputs of post hoc statistical analysis are depicted in Table 5. The obtained results clearly demonstrate the superiority of the OVS decomposition strategy in calculating class potential. By selectively sampling the original dataset in an even proportion, we can preserve the characteristics of each class while accelerating the class potential calculation process and mitigating the dominance of majority classes. The conventional method of calculating pairwise potentials and then averaging them (OVO) is not very satisfactory, potentially due to ignoring the correlation between paired class features. Another method of considering all instances of all classes (OVA) is too slow and tends to be less stable in the presence of strong class imbalance.

After selecting the optimal design strategies for MC-NRO, we conducted the entire experiment and listed the detailed data distribution of multiclass imbalanced datasets in Table 6. The second column is the class distribution of the first fold training data. The third column shows the total number of data and class distribution after oversampling. All the other classes except the largest one will be oversampled, and the oversampling ratio is set to 1. Most of benchmarks will have the same or nearly the same number of samples after oversampling. Refer to the cleaning strategy specific to the multiclass setting (Step 17–22 in Algorithm 2), we clean up the moved majority samples by removing them in the overlapping region after all classes oversampled. The fourth column shows that all the benchmarks have roughly balanced class distribution after the sampling and cleaning process.

#### 4.3. Comparison with reference oversampling methods

In the second stage of the experimental study, we compare the proposed MC-NRO with other advanced oversampling methods to evaluate

<sup>1</sup> <https://github.com/sshiyi77/MC-NRO>.

**Table 6**  
Detailed data distribution of multiclass imbalanced datasets after the sampling and cleaning process.

| Datasets        | 1st fold training data                    | Oversampled training data                             | Cleaned training data                                 |
|-----------------|-------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| Automobile      | 79 (1/10/24/23/14/7)                      | 149 (24/25/24/23/28/25)                               | 126 (24/22/16/18/22/24)                               |
| Balance         | 312 (24/144/144)                          | 427 (139/144/144)                                     | 427 (139/144/144)                                     |
| Car             | 864 (192/35/605/32)                       | 2370 (576/591/605/598)                                | 2167 (530/590/449/598)                                |
| Cleveland       | 148 (80/27/17/18/6)                       | 395 (80/76/80/79/80)                                  | 350 (60/65/72/73/80)                                  |
| Contraceptive   | 736 (314/167/255)                         | 942 (314/314/314)                                     | 836 (251/314/261)                                     |
| Dermatology     | 179<br>(55/30/36/24/24/10)                | 332 (55/60/60/51/52/54)                               | 303 (46/48/55/51/49/54)                               |
| Ecoli           | 168 (71/39/1/1<br>/17/10/3/26)            | 577 (71/78/71/71<br>/74/70/72/70)                     | 496 (50/56/65/65<br>/68/66/70/56)                     |
| Flare           | 533 (74/106/119<br>/47/22/165)            | 991 (166/166/165<br>/165/164/165)                     | 882 (166/147/123<br>/165/164/117)                     |
| Glass           | 107 (35/38/8/7/4/15)                      | 222 (35/38/38/39/38/34)                               | 197 (30/28/32/36/38/33)                               |
| Hayes-roth      | 80 (32/32/16)                             | 96 (32/32/32)                                         | 92 (29/31/32)                                         |
| Led7digit       | 250 (22/19/25/29/26<br>/26/23/29/26/25)   | 300 (30/31/30/29/30<br>/30/31/29/30/30)               | 232 (30/31/23/21/22<br>/20/21/19/20/25)               |
| Lymphogra-phy   | 74 (2/30/41/1)                            | 163 (41/40/41/41)                                     | 136 (41/28/26/41)                                     |
| New-thyroid     | 107 (75/17/15)                            | 222 (75/73/74)                                        | 222 (75/73/74)                                        |
| Page-blocks     | 274 (246/16/2/4/6)                        | 1228 (246/245<br>/246/246/245)                        | 1182 (214/237<br>/246/245/240)                        |
| Thyroid         | 360 (9/18/333)                            | 1000 (334/333/333)                                    | 980 (334/336/310)                                     |
| Vehicle         | 423 (109/106/109/99)                      | 436 (109/109/109/109)                                 | 376 (88/91/89/109)                                    |
| Wine            | 89 (29/36/24)                             | 106 (35/36/35)                                        | 106 (35/36/35)                                        |
| Winequality-red | 799 (5/26/340<br>/319/100/9)              | 2016 (340/338/340<br>/319/341/338)                    | 1752 (320/336/183<br>/258/327/328)                    |
| Yeast           | 742 (231/3/17/22/26<br>/81/122/215/10/15) | 2300<br>(231/231/231/229/234<br>/228/237/215/232/232) | 2148<br>(169/231/231/227/215<br>/218/217/178/231/231) |
| Zoo             | 50 (20/10/2<br>/7/2/4/5)                  | 142 (20/20/20<br>/21/20/21/20)                        | 128 (16/16/20<br>/19/20/19/18)                        |

its practicality. The detailed results for each dataset on C5.0 classifier are presented in Tables 7–9. In Fig. 6, we visualize the results of the win-loss-tie analysis, in which we compare the number of datasets on which MC-NRO is superior, equal or inferior to the reference methods on a pairwise basis. As can be seen, in all of the cases, MC-NRO tends to outperform the reference oversampling methods, which is reflected in the highest average ranks regarding all of evaluation metrics for C5.0 and a majority of wins in a per-dataset pairwise comparison of the methods. Furthermore, the observed performance improvements are statistically significant in comparison to most of the reference methods.

**Performance of neighborhood repartition.** In Section 3.2, we propose to repartition the designed neighborhood to generate synthetic instances within the specific area. The Flare dataset presents a significant analytical challenge, comprising 1066 instances distributed among 6 distinct classes, and its complex structure is characterized by 11 categorical variables (in Table 2). In this case, the proposed MC-NRO restricts the generation process of new instances in the regions with higher class potentials. For Flare, MC-NRO achieves the best results on the three metrics.

**Performance of cleaning strategy.** The class overlap issue can be exacerbated in the multiclass case, as a sample may overlap with multiple classes simultaneously. In Section 3.2, we suggest identifying the moved majority samples near boundary and removes samples in the multiple overlapping regions. Although the Vehicle dataset has the lowest IR, experiments have found that it needs to clean up a large number of overlapping samples after oversampling (in Table 6). MC-NRO outperforms the other algorithms on all three metrics for this dataset.

**Performance of data characteristics.** In Section 3.4, we extend our neighborhood repartition oversampling algorithm to the multiclass setting, we further illustrate its performance with respect to three data characteristics. (1) The Yeast dataset has a **high IR** of 71 (in Table 2). Our proposed MC-NRO is significantly superior to other algorithms on all three metrics for this dataset. (2) Car is the **largest dataset** among all 20 benchmarks. In this case, MC-NRO achieves the best results on AvAcc and mGM metrics, and the second-best results on CBA

metric. (3) Yeast has **small disjuncts**, which consists of three classes containing only two, two and five samples, respectively. Therefore, the class information for small disjuncts is rare. In Section 3.1, we clean up the minority neighborhood by moving the majority class instances outside to provide secure authorization for generating samples. In this instance, MC-NRO outperforms the other algorithms significantly on all three metrics for this dataset.

Among the four classifiers, MC-NRO achieved statistically significantly better performance than all of the reference methods, particularly when compared to the C5.0 and NB classifiers. The C5.0 classifier is more efficient in segmenting and element search, and it does not overfit the data. Since new instances generated by MC-NRO adhere to the density distribution at the base of each class, the NB classifier can also benefit from this. Meanwhile, KNN and MLP depend on a larger number of training instances, and their classification performance may decline on certain datasets. This can be mitigated by using MC-NRO to force the generation of additional instances.

#### 4.4. Evaluation of the impact of class label noise

To evaluate the impact of label noise on the predictive performance of MC-NRO, we introduce random noise labels by substituting the labels of a training subset with other randomly chosen labels at specified noise levels. In our experimental study, the noise level is confined to {0.05, 0.1, 0.15, 0.20, 0.25}.

The experimental results are displayed in Fig. 7 and Table 10. As we can see, with the increase of noise level, MC-NRO generally outperforms existing methods across most datasets. It is worth analyzing how the prediction performance degrades with the increase of noise level. The performance of most multiclass sampling methods drops sharply after exceeding 15% label noise. Although the prediction performance of MC-NRO also decreases, it is not so violent. Furthermore, in terms of adaptability with different classifiers, MC-NRO consistently surpasses other multiclass oversampling methods across most datasets and noise levels. At a noise level of 5%, MC-NRO performs optimally with C5.0 and KNN classifiers, and at 25% noise, it is effective with all evaluated classifiers.



**Fig. 6.** Comparison of MC-NRO with reference oversampling methods for four base classifiers with respect to the number of datasets on which MC-NRO performs better (green), equal (yellow) or worse (red) on 20 datasets.



**Table 7**

Results according to AvAcc [%] for MC-NRO and reference oversampling methods on C5.0 classifier.

| Datasets        | SMOTE-all    | SMOTE-IPF | MDO   | SMOM         | MC-RBO       | MC-CCR       | NROMM        | MC-NRO       |
|-----------------|--------------|-----------|-------|--------------|--------------|--------------|--------------|--------------|
| Automobile      | 80.12        | 75.32     | 78.13 | 79.04        | 77.11        | 76.98        | <b>92.20</b> | 86.16        |
| Balance         | 55.06        | 54.26     | 57.70 | 59.52        | <u>85.32</u> | 82.87        | 81.17        | <b>85.70</b> |
| Car             | 89.84        | 90.96     | 93.36 | 95.18        | 96.43        | <u>97.12</u> | 95.38        | <b>98.73</b> |
| Cleveland       | 28.92        | 24.98     | 28.92 | 28.01        | 35.01        | 37.88        | <u>38.59</u> | <b>39.19</b> |
| Contraceptive   | 50.63        | 52.88     | 53.27 | <b>55.09</b> | 51.93        | 53.18        | <u>54.47</u> | 54.01        |
| Dermatology     | 95.72        | 92.18     | 97.48 | <b>99.31</b> | 96.11        | 94.29        | 97.73        | <u>98.12</u> |
| Ecoli           | 64.68        | 60.43     | 61.16 | 61.16        | 72.06        | <u>74.07</u> | 73.96        | <u>76.17</u> |
| Flare           | <u>71.86</u> | 68.55     | 68.72 | 70.64        | 70.63        | 68.92        | 71.81        | <b>73.37</b> |
| Glass           | 71.52        | 67.52     | 70.64 | 71.55        | 71.86        | 69.77        | <b>80.11</b> | <u>79.24</u> |
| Hayes-roth      | 86.45        | 89.74     | 87.33 | 90.06        | 91.88        | 92.11        | <u>92.70</u> | <b>93.64</b> |
| Led7digit       | 70.48        | 72.39     | 75.03 | <u>75.94</u> | 71.53        | 70.48        | 75.76        | <b>76.23</b> |
| Lymphography    | 73.02        | 74.20     | 76.54 | 74.72        | 81.22        | 79.60        | <b>89.08</b> | 86.17        |
| New-thyroid     | 94.70        | 93.05     | 92.06 | 90.24        | 95.76        | 96.18        | <u>97.78</u> | <b>97.89</b> |
| Page-blocks     | 75.83        | 74.20     | 78.47 | 77.56        | 83.71        | 83.71        | <b>90.90</b> | <u>88.57</u> |
| Thyroid         | 80.02        | 78.91     | 79.14 | 80.96        | 89.43        | 80.52        | <b>96.08</b> | <u>93.67</u> |
| Vehicle         | 73.49        | 71.02     | 70.85 | 70.85        | <u>79.48</u> | 72.71        | 74.59        | <b>80.61</b> |
| Wine            | 92.53        | 90.16     | 93.41 | 93.41        | <u>97.57</u> | 95.28        | 97.14        | <b>97.76</b> |
| Winequality-red | 37.41        | 36.28     | 40.05 | 42.78        | 44.12        | <b>46.93</b> | 45.99        | <u>46.52</u> |
| Yeast           | 51.03        | 53.77     | 54.55 | 56.37        | <u>60.39</u> | 58.39        | 59.20        | <b>61.90</b> |
| Zoo             | 82.61        | 67.3      | 79.09 | 79.09        | 88.56        | 85.92        | <b>97.14</b> | <u>96.93</u> |
| Avg. rank       | 6.05         | 7.30      | 5.70  | 4.80         | 3.75         | 4.25         | 2.30         | 1.50         |

Notes: The best results are highlighted in **bold**, and the second best results are underlined.**Table 8**

Results according to CBA [%] for MC-NRO and reference oversampling methods on C5.0 classifier.

| Datasets        | SMOTE-all | SMOTE-IPF | MDO   | SMOM         | MC-RBO       | MC-CCR       | NROMM        | MC-NRO       |
|-----------------|-----------|-----------|-------|--------------|--------------|--------------|--------------|--------------|
| Automobile      | 54.47     | 70.84     | 75.11 | 73.35        | 80.48        | 77.93        | <b>87.40</b> | <u>83.64</u> |
| Balance         | 45.79     | 57.29     | 57.70 | 60.34        | <u>69.12</u> | 64.92        | 62.35        | <b>69.53</b> |
| Car             | 85.25     | 87.51     | 95.73 | <b>98.37</b> | 96.53        | 95.87        | 93.12        | <u>97.02</u> |
| Cleveland       | 24.09     | 21.99     | 27.34 | 27.34        | 35.01        | 33.91        | <b>36.41</b> | <u>35.98</u> |
| Contraceptive   | 41.63     | 39.99     | 51.69 | 52.97        | 52.77        | <b>55.01</b> | <u>52.97</u> | 52.95        |
| Dermatology     | 86.30     | 88.92     | 95.64 | 93.88        | 94.22        | <u>96.19</u> | 95.94        | <b>96.23</b> |
| Ecoli           | 56.07     | 54.72     | 63.53 | 61.77        | <u>72.01</u> | 68.33        | 67.51        | <b>73.69</b> |
| Flare           | 58.59     | 60.03     | 59.51 | 61.27        | <u>63.56</u> | 61.85        | 62.57        | <b>66.56</b> |
| Glass           | 60.63     | 61.02     | 70.64 | 69.76        | 75.03        | 69.89        | <u>76.08</u> | <b>77.83</b> |
| Hayes-roth      | 77.48     | 75.52     | 84.96 | 87.62        | 88.64        | 90.03        | <u>91.73</u> | <b>92.45</b> |
| Led7digit       | 63.94     | 70.98     | 78.19 | 79.95        | 75.12        | <b>84.07</b> | 81.28        | <u>82.54</u> |
| Lymphography    | 43.21     | 70.83     | 75.75 | 78.39        | 79.02        | 80.66        | <b>86.09</b> | <u>84.30</u> |
| New-thyroid     | 85.04     | 89.69     | 95.22 | 93.46        | 92.84        | 90.14        | <b>95.94</b> | <u>95.33</u> |
| Page-blocks     | 78.60     | 76.35     | 80.84 | 79.96        | 86.73        | 84.63        | <b>89.83</b> | <u>87.43</u> |
| Thyroid         | 80.58     | 70.46     | 76.77 | 75.01        | 84.11        | 81.99        | <b>94.94</b> | <u>91.93</u> |
| Vehicle         | 62.18     | 71.49     | 72.43 | 72.43        | <u>78.05</u> | 71.74        | 74.49        | <b>79.17</b> |
| Wine            | 84.30     | 87.99     | 91.83 | 90.95        | 94.01        | 92.89        | <u>95.87</u> | <b>95.99</b> |
| Winequality-red | 24.76     | 40.01     | 41.63 | 39.87        | <b>47.66</b> | 40.37        | 46.11        | <u>47.41</u> |
| Yeast           | 46.44     | 49.03     | 53.76 | 55.52        | <u>57.97</u> | 57.83        | 57.33        | <b>58.45</b> |
| Zoo             | 60.83     | 79.05     | 76.72 | 76.72        | 82.51        | 81.52        | <b>93.57</b> | <u>92.24</u> |
| Avg. rank       | 7.60      | 7.10      | 5.15  | 5.05         | 3.20         | 3.65         | 2.40         | 1.65         |

Notes: The best results are highlighted in **bold**, and the second best results are underlined.

#### 4.5. Lessons learned

To summarize the experimental study, we try to answer the research questions posed at the beginning of this section.

Q1: What are the optimal parameters setting for MC-NRO, and how they affect the behavior of the algorithm?

MC-NRO is a strongly parameterized preprocessing method, and its predictive performance depends on the correct parameter setting. Based on experimental research, it can be seen that the cleaning strategy is vital to the algorithm, which has the most significant impact on the quality of MC-NRO. The optimal parameter setting for MC-NRO seems to be (i) cleaning by translation and removal, (ii) proportional seed instances selection, and (iii) using the OVS decomposition strategy.

Q2: What is the prediction performance of MC-NRO in comparison to the state-of-the-art oversampling methods?

MC-NRO tends to outperform the state-of-the-art oversampling methods, which is reflected in the highest average ranks regarding all of evaluation metrics and significant differences with other methods.

Q3: How flexible is MC-NRO to be effectively used for various different classifiers?

From the experiments on all considered classifiers, we can see that MC-NRO is highly effective on both noisy and noiseless datasets, particularly with C5.0 and NB classifiers. The other two classifiers still obtain good results.

Q4: How robust is MC-NRO in dealing with label noise?

MC-NRO exhibits remarkable robustness to label noise. As the level of label noise increases or as more classes are affected by noise, its prediction performance declines least and resembles a linear trend. It is particularly noteworthy that MC-NRO is statistically significantly superior to other reference methods, especially for high noise levels (higher than 10%).

**Table 9**

Results according to mGM [%] for MC-NRO and reference oversampling methods on C5.0 classifier.

| Datasets        | SMOTE-all | SMOTE-IPF | MDO   | SMOM  | MC-RBO       | MC-CCR       | NROMM        | MC-NRO       |
|-----------------|-----------|-----------|-------|-------|--------------|--------------|--------------|--------------|
| Automobile      | 51.86     | 73.28     | 75.11 | 78.75 | 77.20        | 75.68        | <b>91.78</b> | 85.60        |
| Balance         | 40.57     | 55.69     | 55.92 | 58.65 | 61.93        | 62.22        | <b>66.74</b> | <u>64.39</u> |
| Car             | 81.16     | 90.85     | 95.73 | 94.82 | <u>97.15</u> | 95.03        | 94.46        | <b>98.45</b> |
| Cleveland       | 18.00     | 23.39     | 26.45 | 24.63 | 34.04        | 30.98        | <b>36.02</b> | <u>34.53</u> |
| Contraceptive   | 38.15     | 50.07     | 51.69 | 52.60 | 50.99        | 49.72        | <u>52.79</u> | <b>52.85</b> |
| Dermatology     | 79.34     | 93.02     | 92.08 | 92.08 | 95.72        | 94.88        | <u>96.42</u> | <b>96.69</b> |
| Ecoli           | 51.72     | 62.88     | 63.53 | 63.53 | <u>72.66</u> | 70.98        | <u>71.72</u> | <b>73.84</b> |
| Flare           | 55.98     | 57.29     | 58.62 | 60.44 | <u>65.50</u> | 64.71        | 64.04        | <b>66.27</b> |
| Glass           | 55.41     | 66.77     | 67.97 | 67.97 | 73.08        | 71.06        | <u>79.74</u> | <b>80.69</b> |
| Hayes-roth      | 70.52     | 84.20     | 83.18 | 86.82 | 81.93        | 85.16        | <b>91.16</b> | <u>88.12</u> |
| Led7digit       | 57.85     | 75.48     | 78.19 | 77.28 | 75.02        | <b>80.44</b> | 78.60        | <u>79.70</u> |
| Lymphography    | 40.60     | 71.86     | 73.08 | 73.99 | 79.13        | 77.36        | <b>88.22</b> | <u>84.29</u> |
| New-thyroid     | 80.69     | 92.55     | 92.55 | 93.46 | 96.73        | 92.17        | <u>97.73</u> | <b>97.96</b> |
| Page-blocks     | 71.64     | 76.72     | 79.06 | 77.24 | 87.27        | 82.55        | <u>88.20</u> | <b>88.37</b> |
| Thyroid         | 73.62     | 75.10     | 75.88 | 77.10 | 86.94        | 81.48        | <b>96.05</b> | <u>93.10</u> |
| Vehicle         | 59.57     | 69.37     | 70.65 | 70.65 | 70.47        | 70.35        | <u>71.68</u> | <b>73.43</b> |
| Wine            | 80.82     | 90.08     | 90.05 | 89.14 | 91.73        | 92.87        | <b>97.05</b> | <u>95.12</u> |
| Winequality-red | 17.80     | 38.42     | 40.74 | 42.56 | <u>48.33</u> | 46.66        | 47.89        | <b>48.94</b> |
| Yeast           | 43.83     | 50.03     | 51.09 | 52.91 | <u>58.06</u> | 56.91        | 55.85        | <b>59.13</b> |
| Zoo             | 55.61     | 78.55     | 75.83 | 77.65 | 85.01        | 84.29        | <b>96.86</b> | <u>93.66</u> |
| Avg. rank       | 8.00      | 6.40      | 5.35  | 4.90  | 3.40         | 4.10         | 2.15         | 1.45         |

Notes: The best results are highlighted in **bold**, and the second best results are underlined.**Table 10**Shaffer's tests for comparison between MC-NRO and reference oversampling methods on two levels of class noise (smallest and largest noise ratios) with respect to each classifier and metric. We report obtained  $p$ -values. Symbol '>' stands for situation in which MC-NRO is statistically superior and '=' for when there are no significant differences.

| Classifier | Hypothesis    | AvAcc     |           | CBA       |           | mGM       |           |
|------------|---------------|-----------|-----------|-----------|-----------|-----------|-----------|
|            |               | 5% noise  | 25% noise | 5% noise  | 25% noise | 5% noise  | 25% noise |
| C5.0       | vs. SMOTE-IPF | >(0.0216) | >(0.0189) | >(0.0153) | >(0.0118) | >(0.0114) | >(0.0031) |
|            | vs. MDO       | >(0.0364) | >(0.0139) | >(0.0316) | >(0.0144) | >(0.0459) | >(0.0193) |
|            | vs. SMOM      | >(0.0385) | >(0.0138) | >(0.0351) | >(0.0154) | >(0.0346) | >(0.0069) |
|            | vs. MC-RBO    | >(0.0418) | >(0.0299) | =(0.0550) | >(0.0195) | >(0.0413) | >(0.0192) |
|            | vs. MC-CCR    | >(0.0408) | >(0.0202) | =(0.0536) | >(0.0187) | >(0.0405) | >(0.0182) |
|            | vs. NROMM     | =(0.0628) | >(0.0304) | =(0.0765) | >(0.0403) | =(0.0521) | >(0.0211) |
| MLP        | vs. SMOTE-IPF | >(0.0243) | >(0.0199) | >(0.0178) | >(0.0121) | >(0.0230) | >(0.0195) |
|            | vs. MDO       | =(0.0887) | >(0.0359) | =(0.0636) | >(0.0354) | =(0.0853) | >(0.0408) |
|            | vs. SMOM      | =(0.1169) | >(0.0439) | =(0.0967) | >(0.0402) | =(0.0986) | >(0.0396) |
|            | vs. MC-RBO    | =(0.2622) | >(0.0482) | =(0.0974) | >(0.0307) | =(0.1069) | >(0.0324) |
|            | vs. MC-CCR    | =(0.1263) | >(0.0472) | =(0.0870) | >(0.0289) | =(0.0955) | >(0.0218) |
|            | vs. NROMM     | =(0.2810) | >(0.0472) | =(0.1024) | >(0.0370) | =(0.1157) | >(0.0468) |
| KNN        | vs. SMOTE-IPF | >(0.0150) | >(0.0069) | >(0.0129) | >(0.0040) | >(0.0118) | >(0.0051) |
|            | vs. MDO       | >(0.0320) | >(0.0158) | >(0.0343) | >(0.0152) | >(0.0156) | >(0.0118) |
|            | vs. SMOM      | >(0.0496) | >(0.0217) | >(0.0353) | >(0.0171) | >(0.0331) | >(0.0186) |
|            | vs. MC-RBO    | >(0.0486) | >(0.0482) | =(0.0642) | >(0.0225) | >(0.0310) | >(0.0168) |
|            | vs. MC-CCR    | >(0.0467) | >(0.0357) | =(0.0549) | >(0.0153) | >(0.0382) | >(0.0112) |
|            | vs. NROMM     | =(0.0579) | >(0.0252) | =(0.0655) | >(0.0306) | =(0.0546) | >(0.0213) |
| NB         | vs. SMOTE-IPF | >(0.0113) | >(0.0091) | >(0.0144) | >(0.0071) | >(0.0138) | >(0.0043) |
|            | vs. MDO       | =(0.0756) | >(0.0428) | =(0.0607) | >(0.0262) | =(0.0714) | >(0.0401) |
|            | vs. SMOM      | =(0.1049) | >(0.0413) | =(0.0579) | >(0.0318) | =(0.0614) | >(0.0394) |
|            | vs. MC-RBO    | =(0.1702) | >(0.0435) | =(0.1003) | >(0.0406) | =(0.1097) | >(0.0321) |
|            | vs. MC-CCR    | =(0.0926) | >(0.0473) | =(0.0798) | >(0.0343) | =(0.0853) | >(0.0287) |
|            | vs. NROMM     | =(0.2208) | >(0.0445) | =(0.1690) | >(0.0337) | =(0.1897) | >(0.0435) |

## 5. Conclusion and future works

The aim of this paper is to propose a novel and effective oversampling method to address the challenge of multiclass imbalanced data classification. We have developed the *Multiclass Neighborhood Repartition-based oversampling* (MC-NRO) algorithm, which uses a selection strategy of oversampled area based on the nearest neighbors and combines the data cleaning operation. Utilizing the neighborhood repartition function, this method can consider the original data characteristics of each class to precisely optimize the placement of new instances and reconstruct the feature space of the minority, which expand decision boundary while suppressing label noise. Experimental

studies on benchmark datasets verify the effectiveness of the proposed solution. MC-NRO has significant advantages over state-of-the-art methods, especially its flexibility in multiple classifiers and its robustness against label noise. It is worth mentioning that the experiment is a progress in using non-depth learning to process multiple types of imbalanced data, and the estimated computational complexity is also acceptable. The results obtained in this paper encourage us to further explore alternative methods for oversampling imbalanced data. Future research may include:

- Selection of the neighborhood of minority classes, which can consider embedding other forms of selection into MC-NRO.

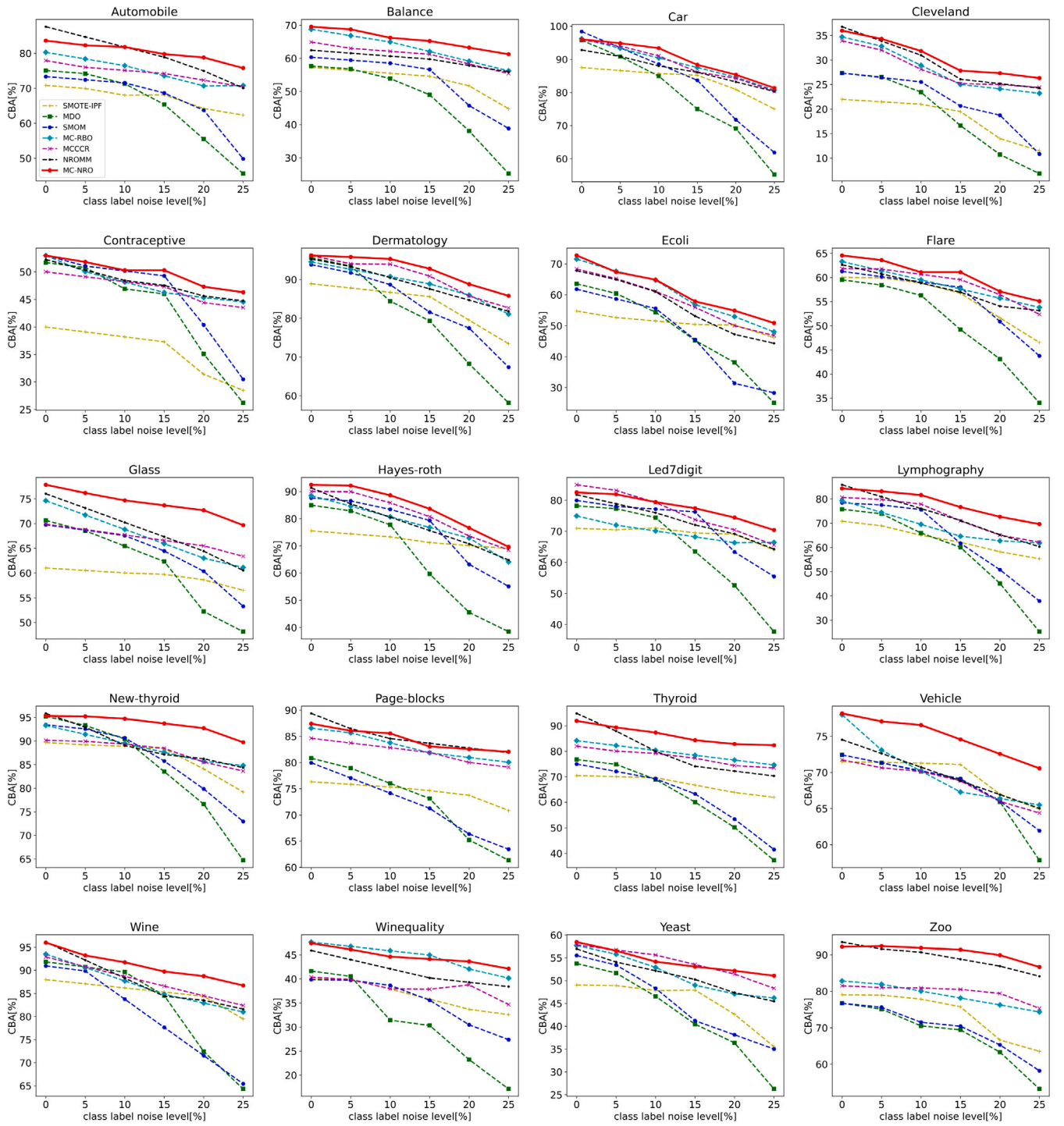


Fig. 7. Influence of varying class label noise levies on MC-NRO and reference oversampling methods according to CBA [%] with C5.0 classifier.

- Propositions of new idea of cleaning the majority observations nearby the minority instances.
- Application of MC-NRO to a real-world imbalanced data susceptible to the presence of label noise, i.e., medical data.

**Yongsong Wang:** Data curation. **Ying Chen:** Project administration, Funding acquisition.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### CRedit authorship contribution statement

**Shiyi Shen:** Writing – original draft, Validation, Methodology, Formal analysis. **Zhixin Li:** Resources, Investigation, Conceptualization. **Zhan Huan:** Writing – review & editing. **Fanqi Shang:** Software.

## Data availability

Implementation and reproducibility. The proposed MC-NRO algorithm was implemented in Python programming language and published as an open-source code at <https://github.com/sshiyi77/MC-NRO>.

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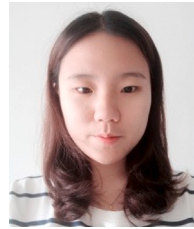
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## References

- [1] D. Li, C. Zheng, J. Zhao, Y. Liu, Diagnosis of heart failure from imbalance datasets using multi-level classification, *Biomed. Signal Process. Control* 81 (2023) 104538, <http://dx.doi.org/10.1016/j.bspc.2022.104538>, URL <https://www.sciencedirect.com/science/article/pii/S1746809422009922>.
- [2] G.-H. Fu, J.-B. Wang, W. Lin, An adaptive loss backward feature elimination method for class-imbalanced and mixed-type data in medical diagnosis, *Chemometr. Intell. Lab. Syst.* 236 (2023) 104809, <http://dx.doi.org/10.1016/j.chemolab.2023.104809>, URL <https://www.sciencedirect.com/science/article/pii/S016974392300059X>.
- [3] Y.-H. Yoo, U.-H. Kim, J.-H. Kim, Recurrent reconstructive network for sequential anomaly detection, *IEEE Trans. Cybern.* 51 (3) (2021) 1704–1715, <http://dx.doi.org/10.1109/TCYB.2019.2933548>.
- [4] J. Zhang, X. Chen, W.W.Y. Ng, C.S. Lai, L.L. Lai, New appliance detection for nonintrusive load monitoring, *IEEE Trans. Ind. Inform.* 15 (8) (2019) 4819–4829, <http://dx.doi.org/10.1109/TII.2019.2916213>.
- [5] S. Liang, H. Yu, Revealing new therapeutic opportunities through drug target prediction: a class imbalance-tolerant machine learning approach, *Bioinformatics* 36 (16) (2020) 4490–4497.
- [6] X. Gao, Z. Meng, X. Jia, J. Liu, X. Diao, B. Xue, Z. Huang, K. Li, An imbalanced binary classification method based on contrastive learning using multi-label confidence comparisons within sample-neighbors pair, *Neurocomputing* 517 (2023) 148–164, <http://dx.doi.org/10.1016/j.neucom.2022.10.069>, URL <https://www.sciencedirect.com/science/article/pii/S0925231222013510>.
- [7] J. Zhang, T. Wang, W.W. Ng, W. Pedrycz, Ensembling perturbation-based oversamplers for imbalanced datasets, *Neurocomputing* 479 (2022) 1–11, <http://dx.doi.org/10.1016/j.neucom.2022.01.049>, URL <https://www.sciencedirect.com/science/article/pii/S0925231222000698>.
- [8] M. Koziarski, M. Woźniak, B. Krawczyk, Combined cleaning and resampling algorithm for multi-class imbalanced data with label noise, *Knowl.-Based Syst.* 204 (2020) 106223, <http://dx.doi.org/10.1016/j.knsys.2020.106223>, URL <https://www.sciencedirect.com/science/article/pii/S0950705120304330>.
- [9] M. Koziarski, M. Woźniak, CCR: A combined cleaning and resampling algorithm for imbalanced data classification, *Int. J. Appl. Math. Comput. Sci.* 27 (4) (2017) 727–736.
- [10] F.K. Inaba, E.O. Teatini Salles, S. Perron, G. Caporossi, DGR-ELM—distributed generalized regularized ELM for classification, *Neurocomputing* 275 (2018) 1522–1530, <http://dx.doi.org/10.1016/j.neucom.2017.09.090>, URL <https://www.sciencedirect.com/science/article/pii/S0925231217316193>.
- [11] S. Wang, X. Yao, Multiclass imbalance problems: Analysis and potential solutions, *IEEE Trans. Syst. Man Cybern. B* 42 (4) (2012) 1119–1130, <http://dx.doi.org/10.1109/TSMCB.2012.2187280>.
- [12] A. Fernández, S. García, M. Galar, R.C. Prati, B. Krawczyk, F. Herrera, *Learning from Imbalanced Data Sets*, vol. 10, no. 2018, Springer, 2018.
- [13] B. Krawczyk, Learning from imbalanced data: open challenges and future directions, *Prog. Artif. Intell.* 5 (4) (2016) 221–232.
- [14] J.A. Sáez, J. Luengo, J. Stefanowski, F. Herrera, SMOTE-IPF: Addressing the noisy and borderline examples problem in imbalanced classification by a re-sampling method with filtering, *Inform. Sci.* 291 (2015) 184–203, <http://dx.doi.org/10.1016/j.ins.2014.08.051>, URL <https://www.sciencedirect.com/science/article/pii/S0020025514008561>.
- [15] J.A. Sáez, B. Krawczyk, M. Woźniak, Analyzing the oversampling of different classes and types of examples in multi-class imbalanced datasets, *Pattern Recognit.* 57 (2016) 164–178, <http://dx.doi.org/10.1016/j.patcog.2016.03.012>, URL <https://www.sciencedirect.com/science/article/pii/S0031320316001072>.
- [16] N.V. Chawla, K.W. Bowyer, L.O. Hall, W.P. Kegelmeyer, SMOTE: synthetic minority over-sampling technique, *J. Artif. Intell. Res.* 16 (2002) 321–357.
- [17] H. Han, W.-Y. Wang, B.-H. Mao, Borderline-SMOTE: a new over-sampling method in imbalanced data sets learning, in: *International Conference on Intelligent Computing*, Springer, 2005, pp. 878–887.
- [18] C. Bunkhumpornpat, K. Sinapiromsaran, C. Lursinsap, Safe-level-smote: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem, in: *Advances in Knowledge Discovery and Data Mining: 13th Pacific-Asia Conference, PAKDD 2009 Bangkok, Thailand, April 27–30, 2009 Proceedings 13*, Springer, 2009, pp. 475–482.
- [19] Q. Chen, Z.-L. Zhang, W.-P. Huang, J. Wu, X.-G. Luo, PF-SMOTE: A novel parameter-free SMOTE for imbalanced datasets, *Neurocomputing* 498 (2022) 75–88, <http://dx.doi.org/10.1016/j.neucom.2022.05.017>, URL <https://www.sciencedirect.com/science/article/pii/S0925231222005495>.
- [20] M. Koziarski, Potential anchoring for imbalanced data classification, *Pattern Recognit.* 120 (2021) 108114, <http://dx.doi.org/10.1016/j.patcog.2021.108114>, URL <https://www.sciencedirect.com/science/article/pii/S0031320321003010>.
- [21] Y. Liu, Y. Liu, Position-aware safe boundary interpolation oversampling, in: *2020 25th International Conference on Pattern Recognition, ICPR, IEEE, 2021*, pp. 5519–5526.
- [22] M. Koziarski, B. Krawczyk, M. Woźniak, Radial-based oversampling for noisy imbalanced data classification, *Neurocomputing* 343 (2019) 19–33, <http://dx.doi.org/10.1016/j.neucom.2018.04.089>, Learning in the Presence of Class Imbalance and Concept Drift. URL <https://www.sciencedirect.com/science/article/pii/S0925231219301596>.
- [23] F. Li, X. Zhang, X. Zhang, C. Du, Y. Xu, Y.-C. Tian, Cost-sensitive and hybrid-attribute measure multi-decision tree over imbalanced data sets, *Inform. Sci.* 422 (2018) 242–256, <http://dx.doi.org/10.1016/j.ins.2017.09.013>, URL <https://www.sciencedirect.com/science/article/pii/S0020025517304784>.
- [24] A. Fernández, V. López, M. Galar, M.J. del Jesus, F. Herrera, Analysing the classification of imbalanced data-sets with multiple classes: Binarization techniques and ad-hoc approaches, *Knowl.-Based Syst.* 42 (2013) 97–110, <http://dx.doi.org/10.1016/j.knsys.2013.01.018>, URL <https://www.sciencedirect.com/science/article/pii/S0950705113000300>.
- [25] Z.-L. Zhang, X.-G. Luo, S. González, S. García, F. Herrera, DRCW-ASEG: One-versus-one distance-based relative competence weighting with adaptive synthetic example generation for multi-class imbalanced datasets, *Neurocomputing* 285 (2018) 176–187, <http://dx.doi.org/10.1016/j.neucom.2018.01.039>, URL <https://www.sciencedirect.com/science/article/pii/S0925231218300584>.
- [26] B. Krawczyk, M. Koziarski, M. Woźniak, Radial-based oversampling for multiclass imbalanced data classification, *IEEE Trans. Neural Netw. Learn. Syst.* 31 (8) (2020) 2818–2831, <http://dx.doi.org/10.1109/TNNLS.2019.2913673>.
- [27] Y. Liu, Y. Liu, B.X. Yu, S. Zhong, Z. Hu, Noise-robust oversampling for imbalanced data classification, *Pattern Recognit.* 133 (2023) 109008, <http://dx.doi.org/10.1016/j.patcog.2022.109008>, URL <https://www.sciencedirect.com/science/article/pii/S0031320322004885>.
- [28] T. Zhu, Y. Lin, Y. Liu, Synthetic minority oversampling technique for multiclass imbalance problems, *Pattern Recognit.* 72 (2017) 327–340, <http://dx.doi.org/10.1016/j.patcog.2017.07.024>, URL <https://www.sciencedirect.com/science/article/pii/S0031320317302947>.
- [29] L. Yin, Y. Ge, K. Xiao, X. Wang, X. Quan, Feature selection for high-dimensional imbalanced data, *Neurocomputing* 105 (2013) 3–11, <http://dx.doi.org/10.1016/j.neucom.2012.04.039>, Learning for Scalable Multimedia Representation. URL <https://www.sciencedirect.com/science/article/pii/S0925231212007126>.
- [30] L. Abdi, S. Hashemi, To combat multi-class imbalanced problems by means of over-sampling techniques, *IEEE Trans. Knowl. Data Eng.* 28 (1) (2016) 238–251, <http://dx.doi.org/10.1109/TKDE.2015.2458858>.
- [31] G. Collell, D. Prelec, K.R. Patil, A simple plug-in bagging ensemble based on threshold-moving for classifying binary and multiclass imbalanced data, *Neurocomputing* 275 (2018) 330–340, <http://dx.doi.org/10.1016/j.neucom.2017.08.035>, URL <https://www.sciencedirect.com/science/article/pii/S092523121731456X>.
- [32] G. Haixiang, L. Yijing, L. Yanan, L. Xiao, L. Jinling, BPSo-Adaboost-KNN ensemble learning algorithm for multi-class imbalanced data classification, *Eng. Appl. Artif. Intell.* 49 (2016) 176–193, <http://dx.doi.org/10.1016/j.engappai.2015.09.011>, URL <https://www.sciencedirect.com/science/article/pii/S0952197615002110>.
- [33] A. Rodriguez, A. Laio, Clustering by fast search and find of density peaks, *Science* 344 (6191) (2014) 1492–1496.
- [34] X. Zhu, X. Wu, Q. Chen, Eliminating class noise in large datasets, in: *Proceedings of the 20th International Conference on Machine Learning, Icm-03*, 2003, pp. 920–927.
- [35] L.P. Garcia, A.C. de Carvalho, A.C. Lorena, Effect of label noise in the complexity of classification problems, *Neurocomputing* 160 (2015) 108–119, <http://dx.doi.org/10.1016/j.neucom.2014.10.085>, URL <https://www.sciencedirect.com/science/article/pii/S0925231215001241>.
- [36] B. Frenay, M. Verleysen, Classification in the presence of label noise: A survey, *IEEE Trans. Neural Netw. Learn. Syst.* 25 (5) (2014) 845–869, <http://dx.doi.org/10.1109/TNNLS.2013.2292894>.



- [37] W.R. Hendee, To err is human: Building a safer health system, *J. Vasc. Int. Radiol.* 12 (1, Supplement) (2001) P112–P113, [http://dx.doi.org/10.1016/S1051-0443\(01\)70072-3](http://dx.doi.org/10.1016/S1051-0443(01)70072-3), Scientific Session and Scientific Poster Abstracts. URL <https://www.sciencedirect.com/science/article/pii/S1051044301700723>.
- [38] J.C. Chang, S. Amershi, E. Kamar, Revolt: Collaborative crowdsourcing for labeling machine learning datasets, in: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2017, pp. 2334–2346.
- [39] B. Li, Y. Wang, A. Singh, Y. Vorobeychik, Data poisoning attacks on factorization-based collaborative filtering, *Adv. Neural Inf. Process. Syst.* 29 (2016).
- [40] G.E. Batista, R.C. Prati, M.C. Monard, A study of the behavior of several methods for balancing machine learning training data, *ACM SIGKDD Explor. Newsl.* 6 (1) (2004) 20–29.
- [41] H. He, Y. Bai, E.A. Garcia, S. Li, ADASYN: Adaptive synthetic sampling approach for imbalanced learning, in: *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*, Ieee, 2008, pp. 1322–1328.
- [42] J. Derrac, S. Garcia, L. Sanchez, F. Herrera, Keel data-mining software tool: Data set repository, integration of algorithms and experimental analysis framework, *J. Mult. Valued Logic Soft. Comput.* 17 (2015) 255–287.
- [43] P. Cerda, G. Varoquaux, B. Kégl, Similarity encoding for learning with dirty categorical variables, *Mach. Learn.* 107 (8) (2018) 1477–1494.
- [44] P. Branco, L. Torgo, R.P. Ribeiro, Relevance-based evaluation metrics for multi-class imbalanced domains, in: *Advances in Knowledge Discovery and Data Mining: 21st Pacific-Asia Conference, PAKDD 2017, Jeju, South Korea, May 23–26, 2017, Proceedings, Part I 21*, Springer, 2017, pp. 698–710.
- [45] N. Japkowicz, M. Shah, *Evaluating Learning Algorithms: A Classification Perspective*, Cambridge University Press, 2011.
- [46] S. García, A. Fernández, J. Luengo, F. Herrera, Advanced nonparametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power, *Inform. Sci.* 180 (10) (2010) 2044–2064, <http://dx.doi.org/10.1016/j.ins.2009.12.010>, Special Issue on Intelligent Distributed Information Systems. URL <https://www.sciencedirect.com/science/article/pii/S0020025509005404>.



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